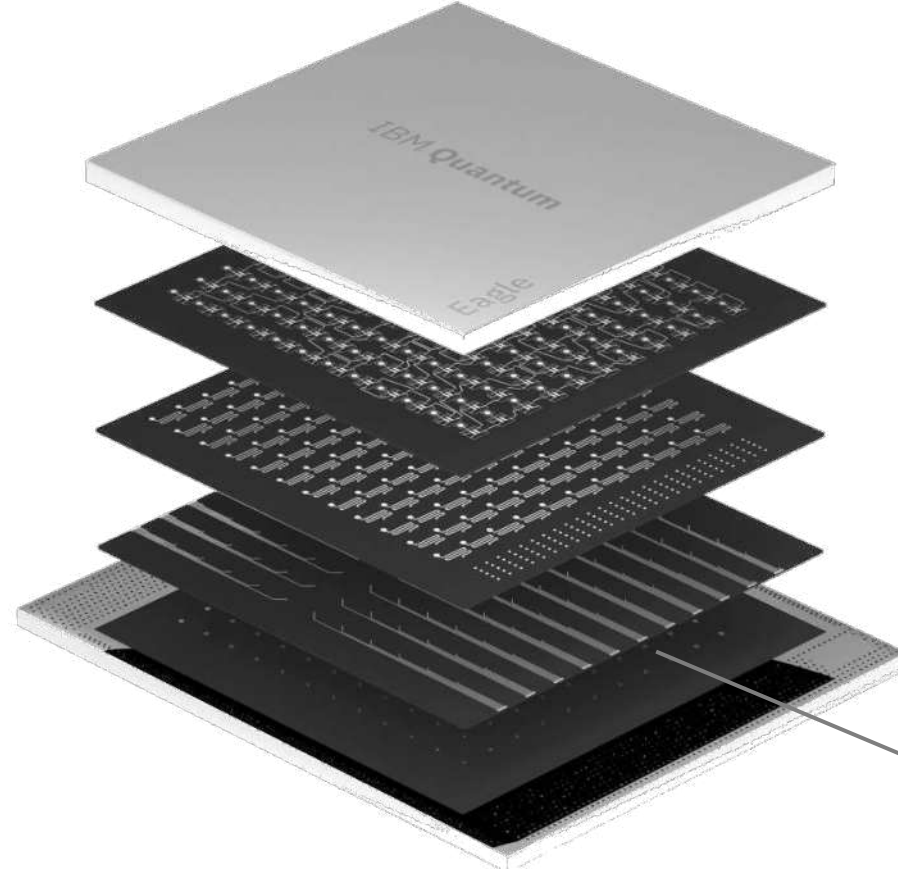


Hello Quantum World!



Zlatko K. Minev, Ph.D.

IBM Quantum, T.J. Watson Research Center, Yorktown Heights, NY



@zlatko_minev

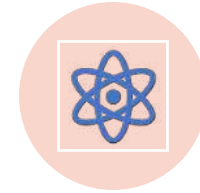


zlatko-minev.com

Hello Quantum World!



About me: Career
overview



Lessons along the
way



Quantum Computers
and IBM Quantum



Experiments on large
devices / making
quantum chips / ...

Avoid firehose of information



Thanks to Fred Moxley for gif reference.

Overview of career in brief

at time $t \approx 0$

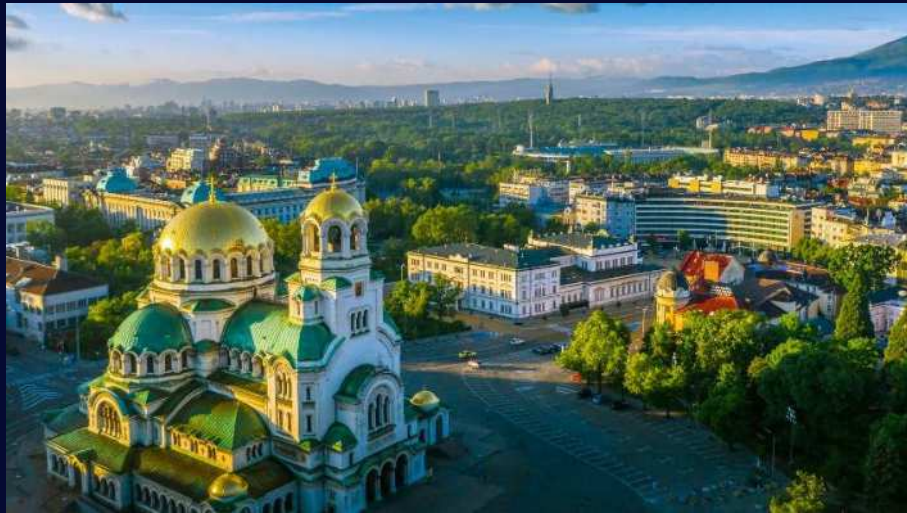
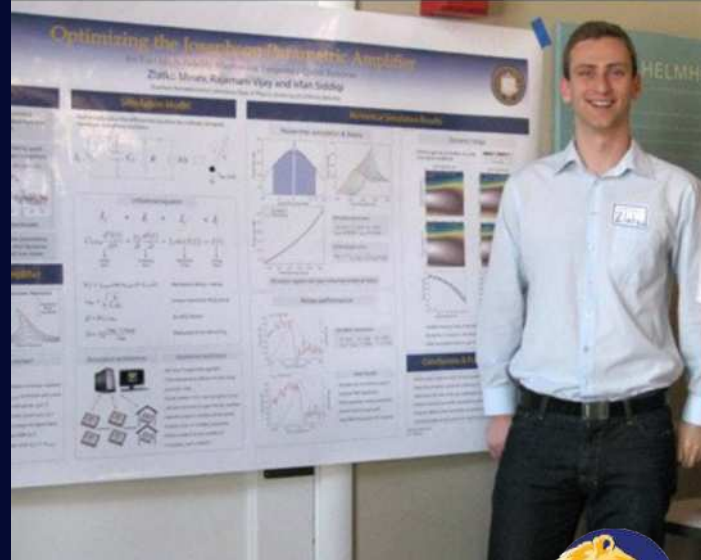
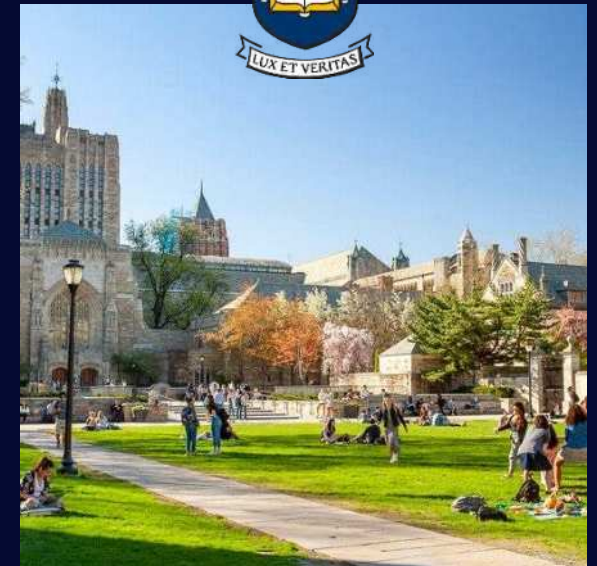


Image: National Geographic

Sofia, Bulgaria



UC Berkeley, CA



Yale University, CT

Roles since then

IBM Quantum

- **Technical Lead and Manager**
for 2 teams and missions I founded:
 - **Qiskit Leap Research**
advanced collaborative quantum computing research
 - **Qiskit Metal**
quantum processor design and analysis

ADDITIONAL ROLES

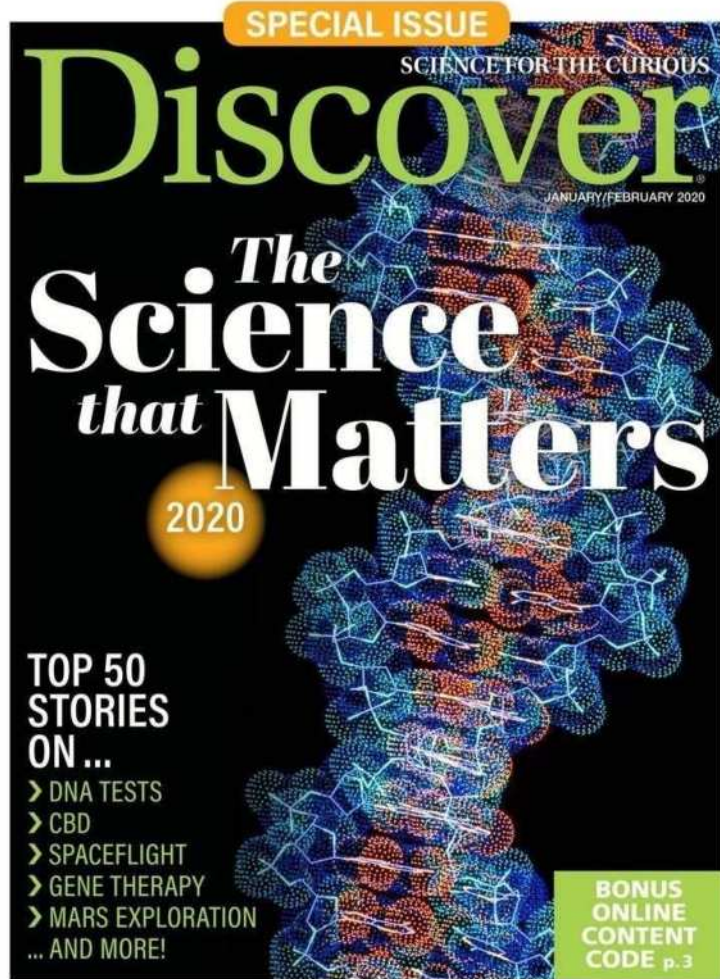
- **Associate Fellow,**
Canadian Institute for Advanced Research (CIFAR)
- **Member of the Board,**
Yale University, Graduate School Alumni Association
- **Board of Governors,**
Yale Alumni Association
- **Lead and Host,**
IBM Quantum, YouTube Seminar
- **Chairman,**
Open Labs Science Outreach



Zlatko Minev



Some highlights





INNOVATORS UNDER 35



Zlatko Minev
AGE: 30
AFFILIATION: IBM QUANTUM RESEARCH, TJ WATSON
COUNTRY OF BIRTH: BULGARIA

His discovery could reduce errors in quantum computing.

In chaotic times it can be reassuring to see so many people working toward a better world. That's true for medical professionals fighting a pandemic and for ordinary citizens fighting for social justice. And it's true for those among us striving to employ technology to address those problems and many others.

Zlatko Minev overturned a mainstay of quantum physics that had troubled Niels Bohr and Albert Einstein alike. For most of the 20th century, it was assumed that atoms change from one energy level to another in abrupt, unpredictable, discrete quantum jumps. Minev proved otherwise.

nature
International journal of science

Letter | Published: 03 June 2019

To catch and reverse a quantum jump mid-flight



Yale-Jefferson Award for Public Service

Forbes

You Don't Have To Be A Rocket (Or Quantum) Scientist To Design A Quantum Computer Chip Using IBM's New Tool Called Qiskit Metal

Paul Smith-Goodson Contributor
Moor Insights and Strategy Contributor Group G
Covid
Global Innovation Observer Contributor

Quantum Jumps



H.J. Carmichael



Letter | Published: 03 June 2019

To catch and reverse a quantum jump mid-flight

Z. K. Minev , S. O. Mundhada, S. Shankar, P. Reinhold, R. Gutiérrez-Jáuregui, R. J. Schoelkopf, M. Mirrahimi, H. J. Carmichael & M. H. Devoret 

Nature **570**, 200–204 (2019) | [Download Citation](#) 



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Discover

SCIENCE FOR THE CURIOUS



Quanta magazine

Economy.bg

Бизнесът в България

b tv



THE QUANTUM DAILY
QUANTUM COMPUTING AND BEYOND

MIT
Technology
Review

...

Zlatko Minev, IBM Quantum

Catching and Reversing a Quantum Jump
Mid-Flight

A Dissertation
Presented to the Faculty of the Graduate School
of
Yale University
in Candidacy for the Degree of
Doctor of Philosophy



by
Zlatko K. Minev

Dissertation Director: Michel H. Devoret

May 2018

Quantum

Discreteness

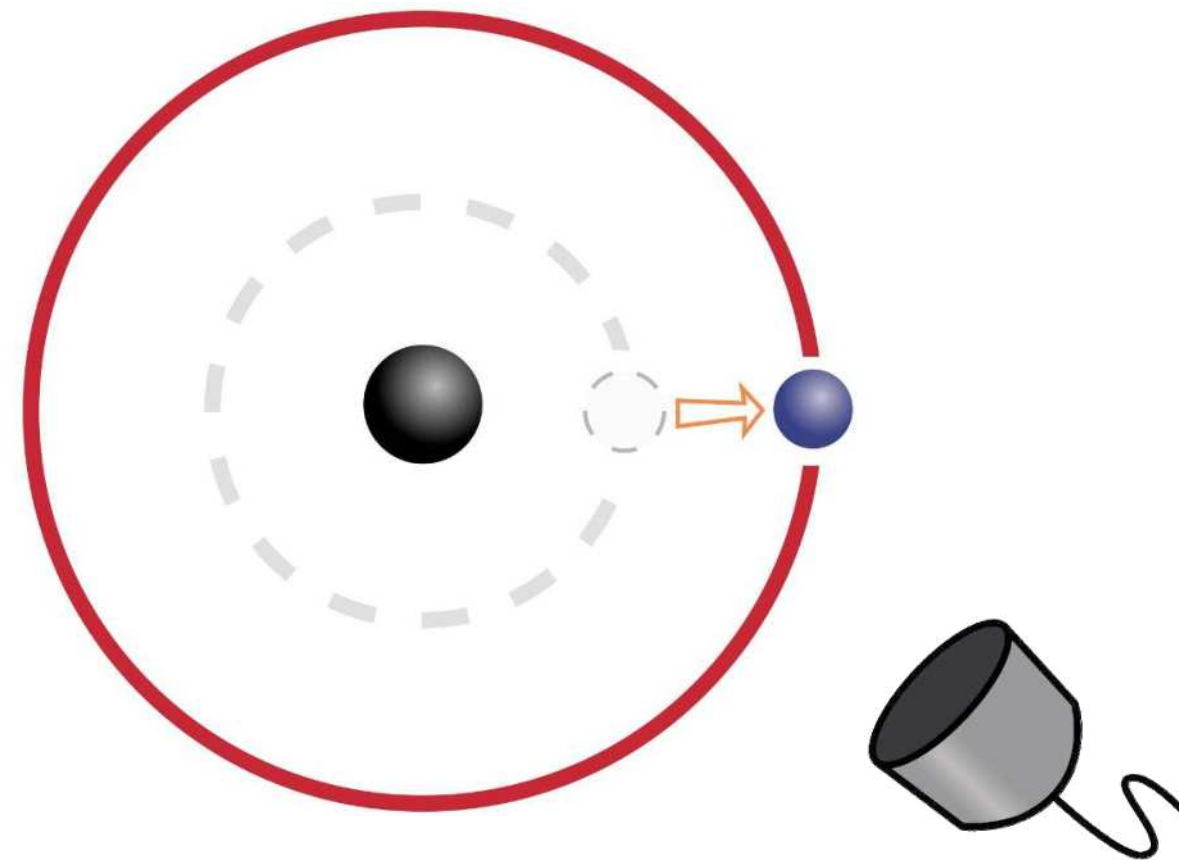
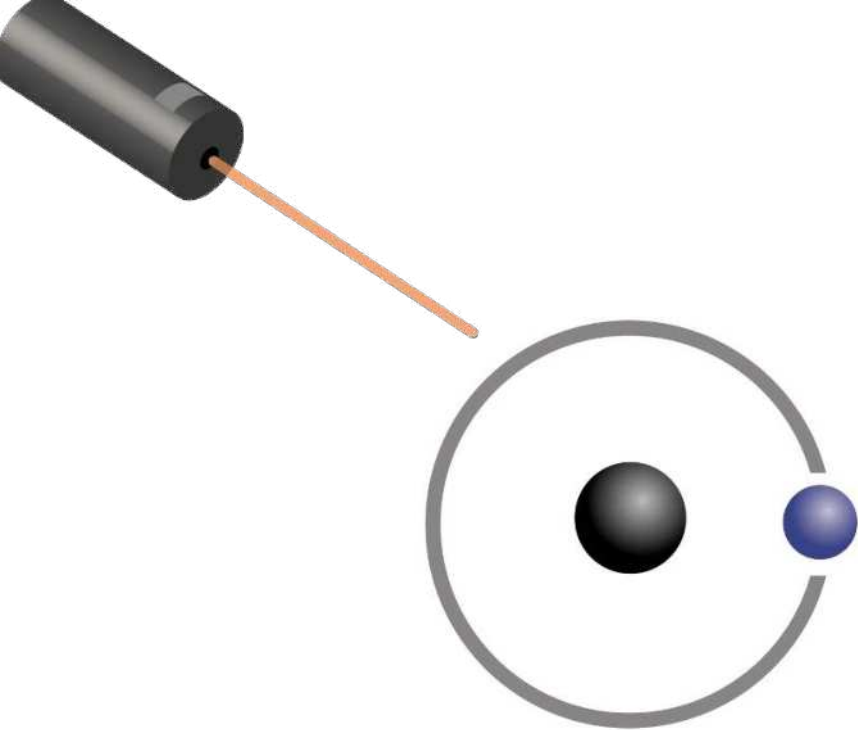


Unpredictability



Predict?

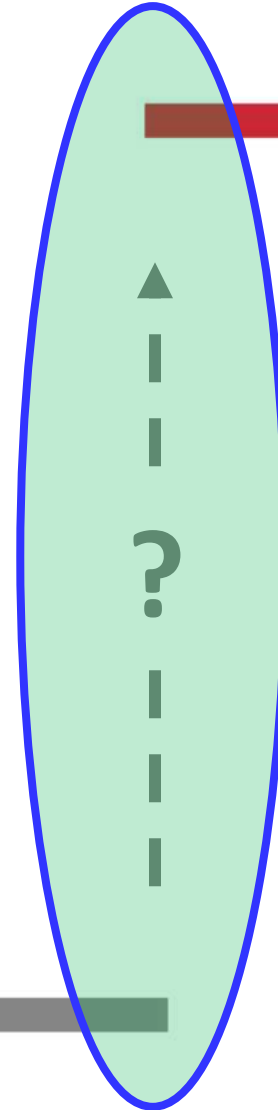




Quantum Jump: Until Now



$|1\rangle$

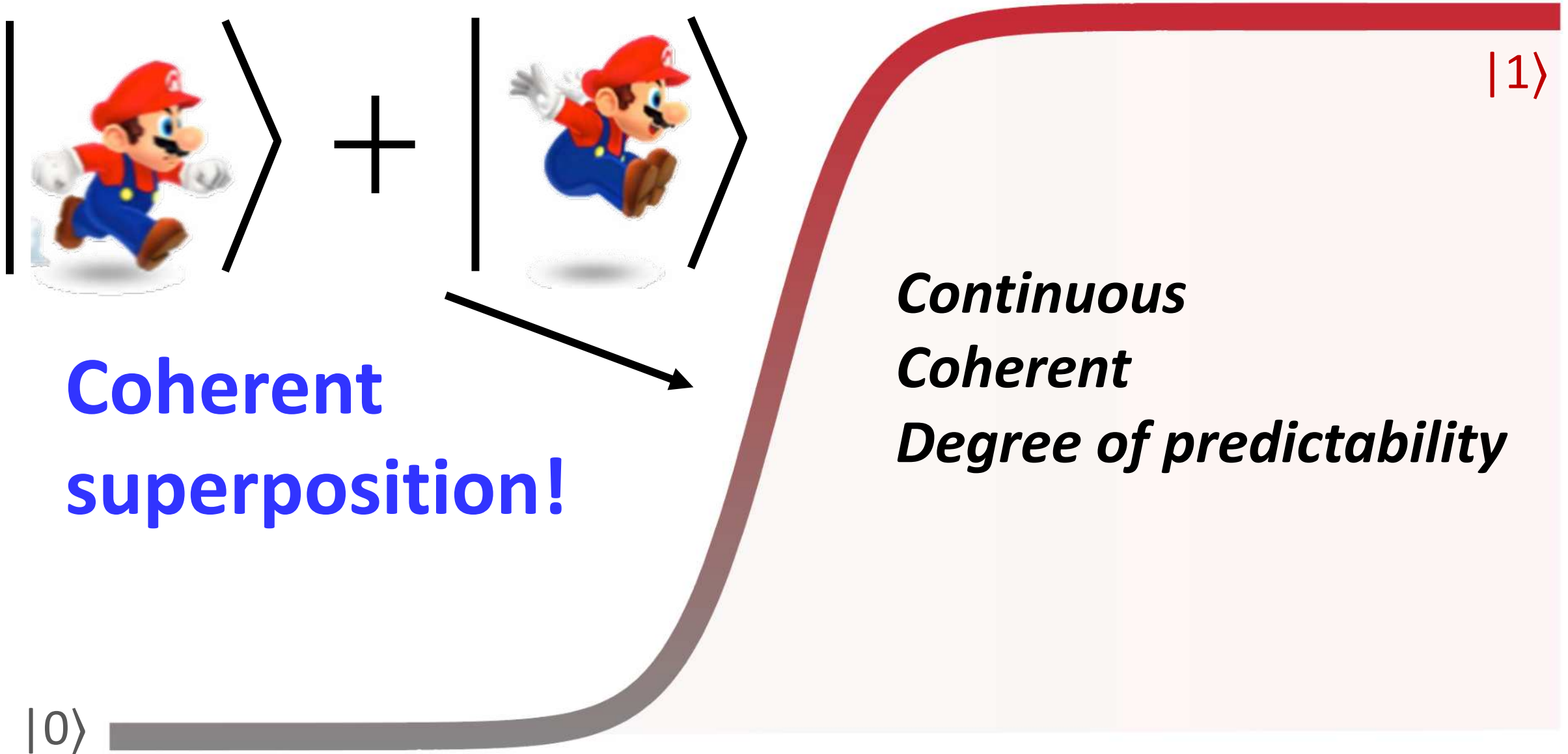


Discrete?
Instantaneous?
Unpredictable?

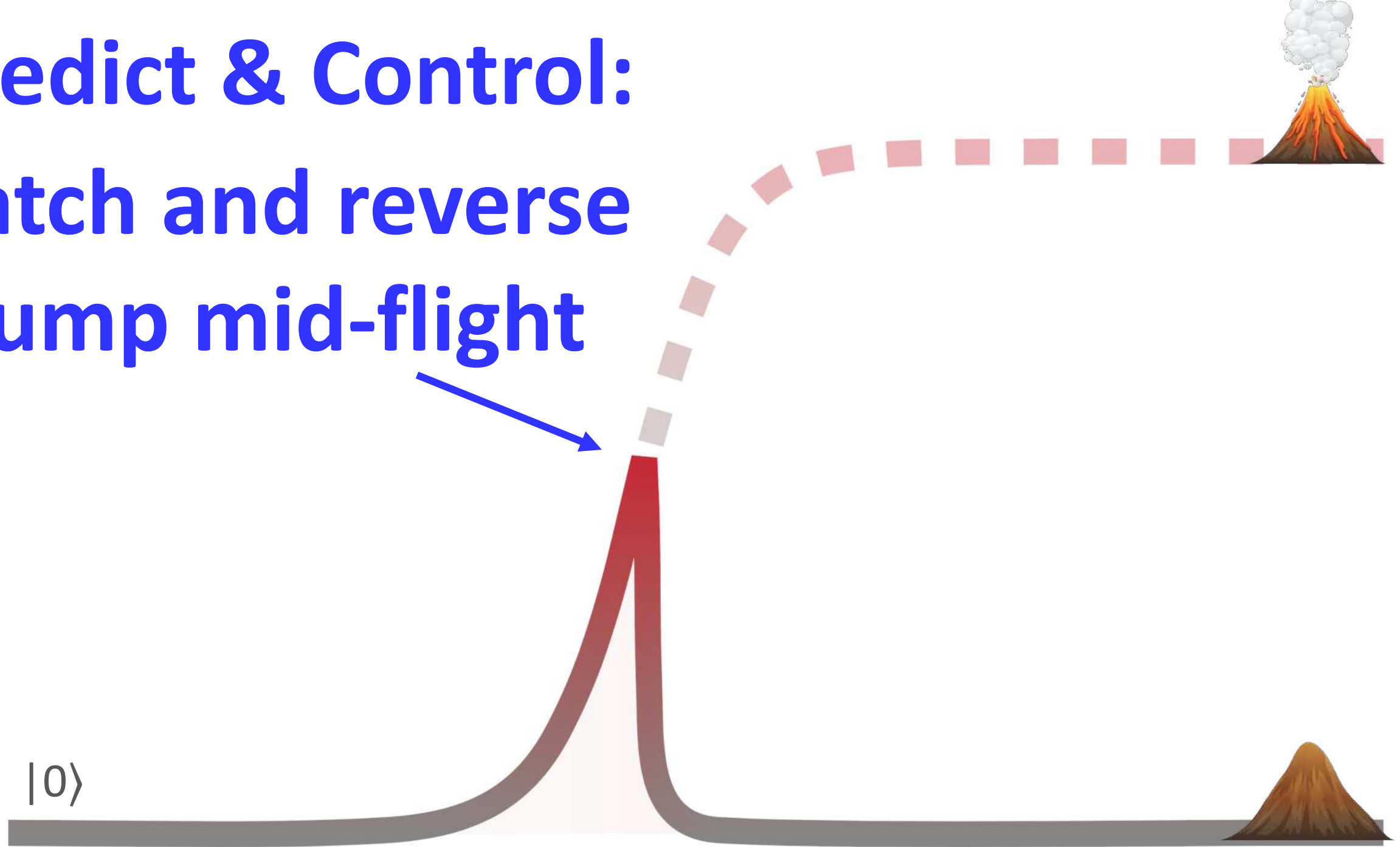
$|0\rangle$



The Real Quantum Jump



Predict & Control: Catch and reverse jump mid-flight



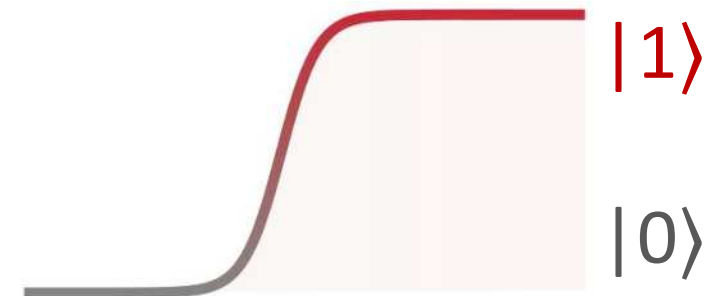
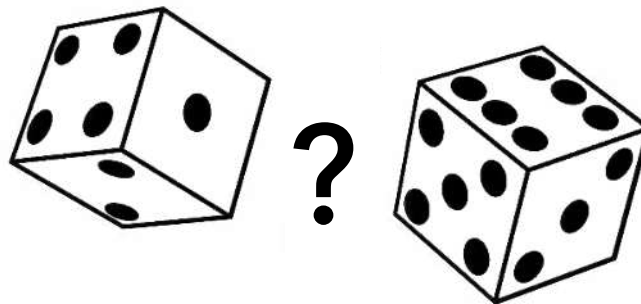
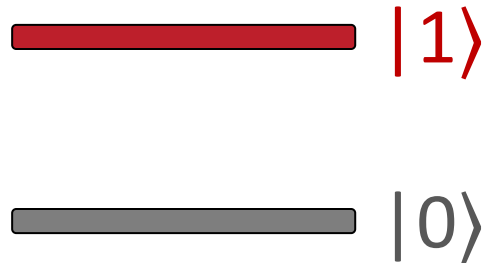
Bohr



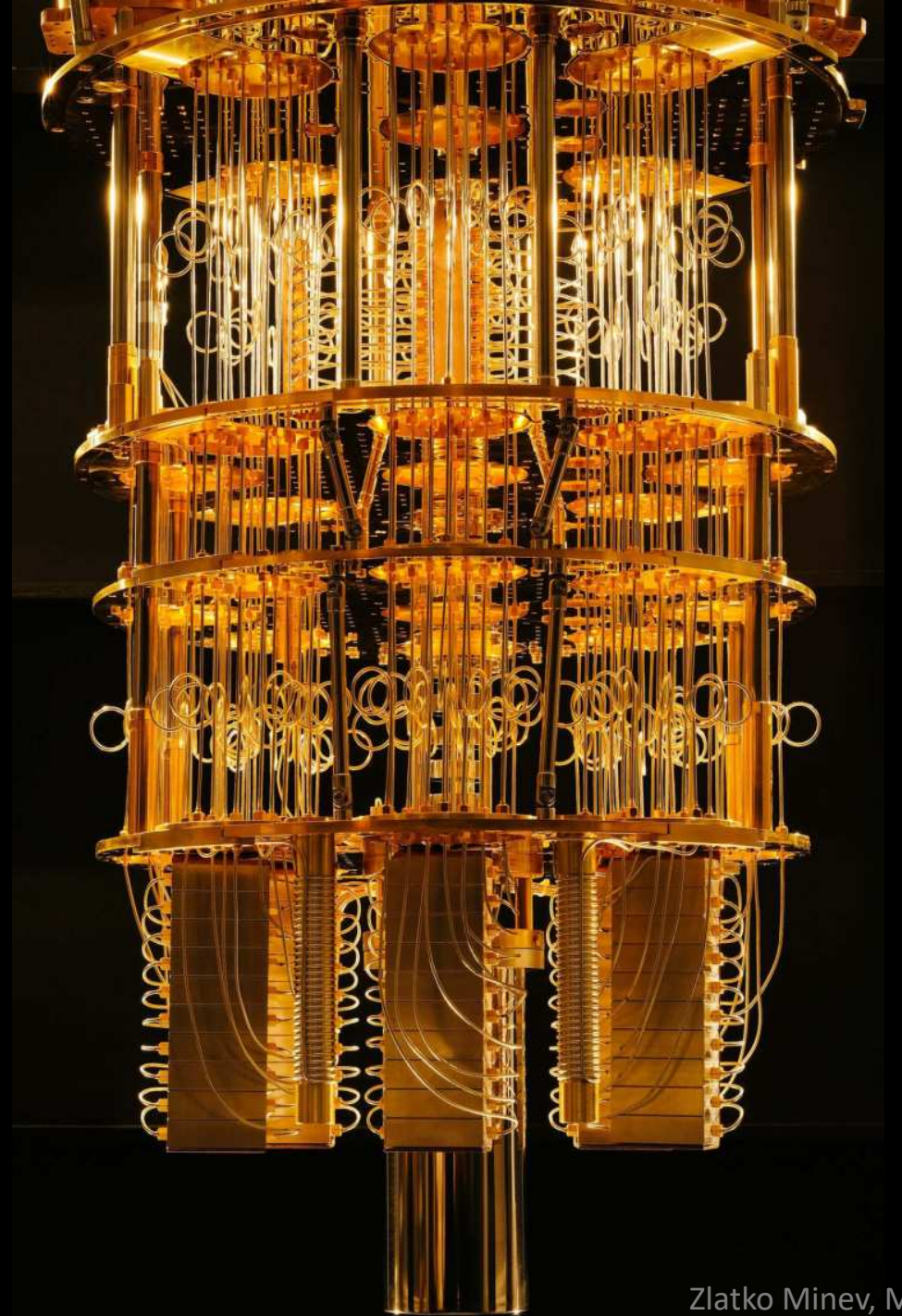
Einstein



Schrödinger



Quantum Computing: Error correction Quantum Sensing



There is more to the story of quantum physics ...

Not quite as discrete and random as we thought ...

Opens doors to otherwise inaccessible technology ...

Lessons

“It is by *logic* that we prove,
but by *intuition* that we discover.”

Henri Poincaré



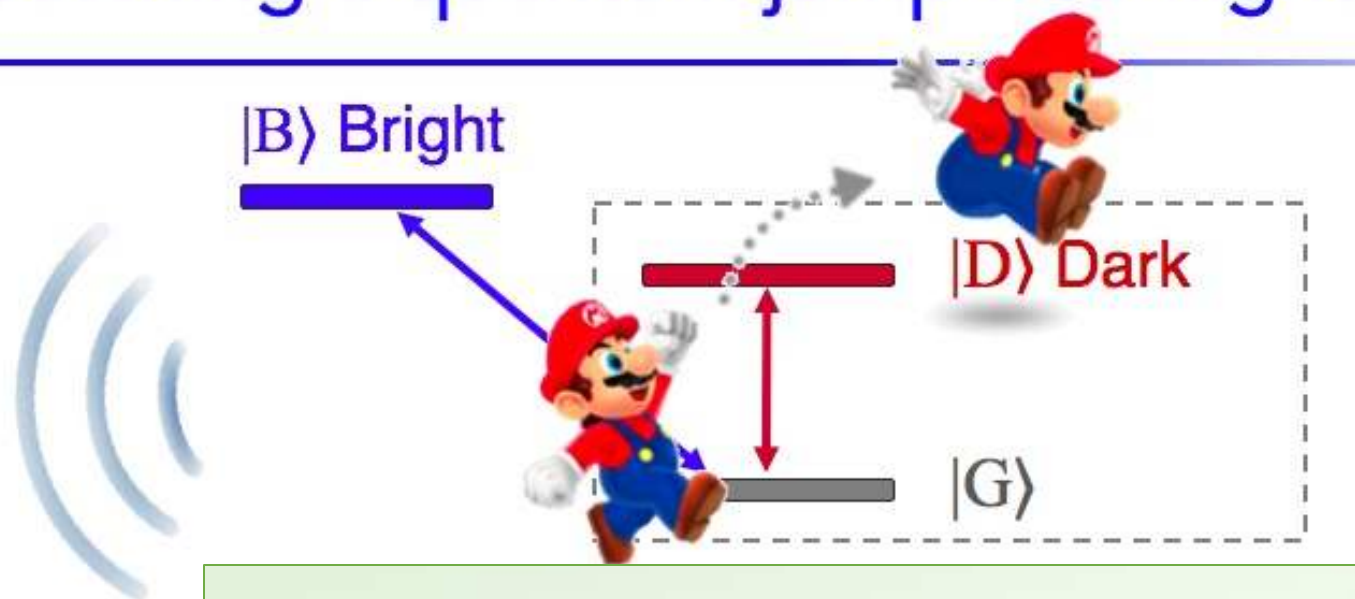
Photo by Eugène Pirou

Believe in your ideas

Hear criticism,
then make a quantum leap



Catching a quantum jump mid-flight



Have fun & misbehave a bit

(John Cohn)

If I knew what I was doing, it
wouldn't be called research.

Albert Einstein
See Hawken *et al.* (2010)

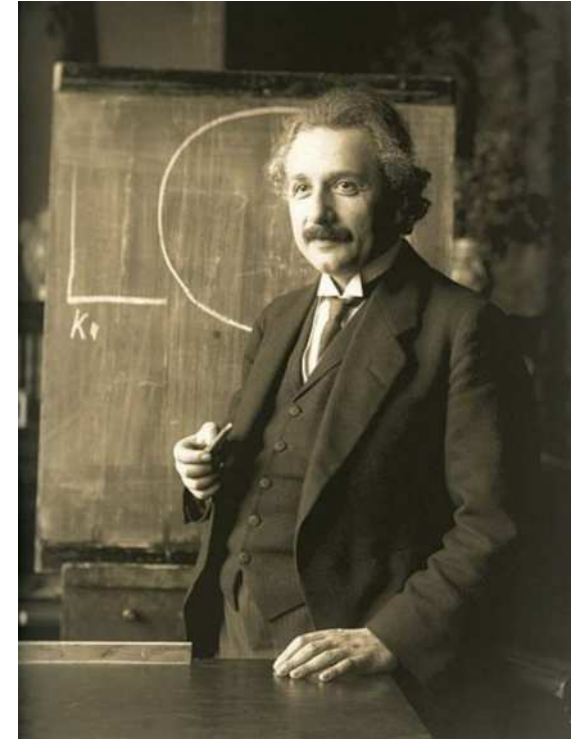


Photo: F. Schmutzer

Learn beyond apparent boundaries

“We are not students of some subject matter, but students of problems. And problems may cut right across the borders of any subject matter or discipline.”

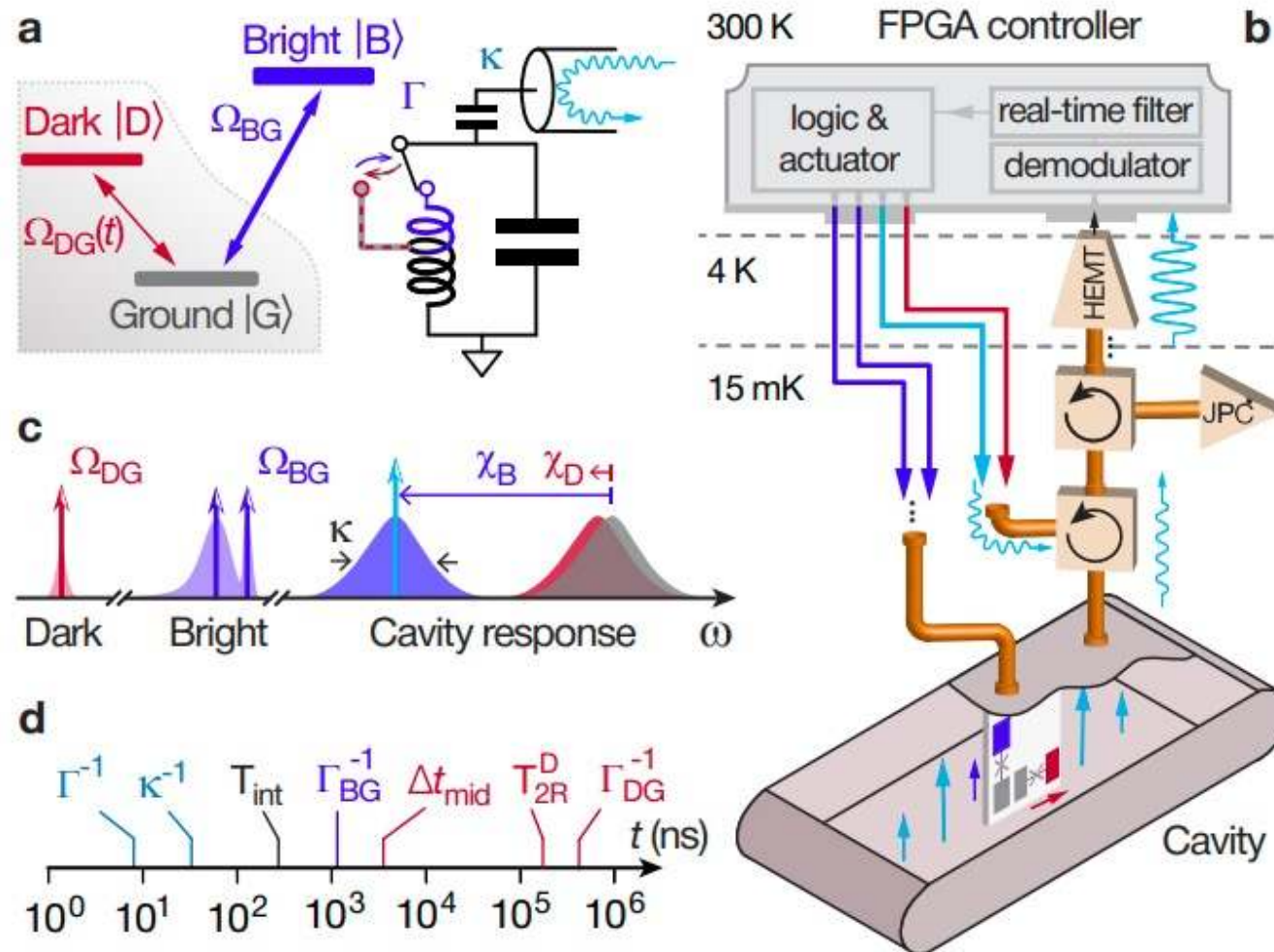
Karl Popper



"I'M ON THE VERGE OF A MAJOR BREAKTHROUGH, BUT I'M ALSO AT THAT POINT WHERE CHEMISTRY LEAVES OFF AND PHYSICS BEGINS, SO I'LL HAVE TO DROP THE WHOLE THING."

We're in the business of challenging

There are no unsolvable problems, only paths which do not lead to their solution



"Having fun can smooth the ride" (John Cohn, IBM Fellow)



Lessons from the quantum jumps story

Question-driven research

You can only have great answers
if you start with great questions

Believe in your ideas

Hear criticism,
then make a quantum leap

We're in the business of challenging

There are no unsolvable problems, only
paths which do not lead to their solution

Learn, learn, learn

Learn widely; learn always; do not limit self;
Can't just keep doing what you've been
doing

Periods of intense single- minded focus

Alternate with learning widely

Have fun & misbehave a bit

(John Cohn)

The science of success & the art of fulfillment



Teach to learn

Find the way,
Go the way,
Show the way

i.e., give back

Extra broader work

Broader work
adjacent to research and tech



Educational, enablement, advocacy



140 years in 115 seminars: Quantum research community connects on YouTube



Results.

So how is it going?

115+ seminar

244,461+ hours of total viewer watch time

1 million + viewers (people who watched the video)

Nearly 26 million impressions (people who saw the video in their feed)

Equivalent to 140 years of a person's work life [i.e., full-time equivalent (FTE)]



researcher onboarding guide,
theorist cheat sheet,
researcher code assets,

Qiskit Cheatsheet

1 Introduction

What is Qiskit Leap?

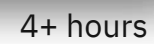
- Forming academic collaborations, our team's mission is to seek and demonstrate the best possible science with quantum computers today.

2 Device Specifications

2.1 Connectivity (IBM Auckland)

Educational assets

Tutorials, videos, manuscripts, ...



Introduction to Quantum Computing and Quantum Hardware

Superconducting circuits

2.1.1 Atomic Energy Levels: Quantization of Energy

From the above discussion, we observe that the atoms can emit light only in discrete bands of frequency. A frequency is thus set corresponded to the gap in energy between two energy levels of the atoms — and not every possible one, only the transition between certain energy levels is allowed. This is referred to as the quantization of energy.

In Figs. 3-5, the potential energy landscape $V(r)$ (shown by the dotted)

Introduction to noise in quantum computers

2 *Oncidium variegatum* (L.) Schlechter

the state $|0\rangle$ as a vector in the Bloch sphere that points from the origin of Bloch sphere all the way to the surface (radius equal to 1) in the $\hat{z}$ direction.

Fig. 22 also shows a black curve along the X -axis. The black curve is the same as the one obtained by the operator X , and corresponding entry $R_X(X)$, will contain a vector zero on the black space.

We now look at the way that the X -gate operates on the initial state $|0\rangle = |0\rangle_1 \otimes |0\rangle_2$. We can see that the ideal X -gate operation will flip $|0\rangle_1$ to $|1\rangle_1$ and $|0\rangle_2$ to $|1\rangle_2$. The physical implementation of an X -gate ($R_X(X)$) can behave as the ideal X -gate (up to an ignorable global phase). An ϵ -approximate $R_X(X)$ operator requires the general state ϕ (control) to have the 1 -bit local state as illustrated in Fig. 24 with the control state ϕ area $1/2$.

Suggesting this instruction with some math, $R_2(\theta)$'s action on $|\psi_0\rangle$ from $\theta = 0$ to $\theta = \pi$, yields the state as

$$\begin{aligned} R_2(\theta) &= \pi(\theta) = i \cos(\pi/2\theta) - i \sin(\pi/2\theta) |0\rangle \\ \cos(\theta = \pi) &= i \cos(\pi/2) - i \sin(\pi/2) |0\rangle \\ &= i \cos(\pi/2) - i \sin(\pi/2) |1\rangle \\ &= -i |1\rangle. \end{aligned}$$

Let's look at the expectation value of the X , Y , and Z quadratures for the state as it evolves as far as angle θ .

$$\begin{aligned} (A) &= (x_1/\theta) \cdot X \cos(\theta) = 0, & (2.7a) \\ (Y) &= (x_2/\theta) \cdot Y \sin(\theta) = -\sin(\theta), & (2.7b) \\ (Z) &= (x_3/\theta) \cdot Z \cos(\theta) = \cos(\theta). & (2.7c) \end{aligned}$$

The time expectation values match the view of the flash sphere. The Δ expectation value is zero the entire excitation, and thus its modulation $-\sin(\theta)$ and $\cos(\theta)$ along Y and Z axis respectively as θ increases. Note we can see the reason as to why the factor of $1/2$ in the definition of the

Of course, you can continue this angle θ under the $R_X(\theta)$ operation. You'll notice that for increasing the angle of θ , it will trace a full circle of the way around the Bloch sphere, back into the ground state when $\theta = 2\pi$.

Fig. 1-4. Black spheres denote the axial

more and more X gets, the more will continue to rotate itself. This visual aid helps to keep in mind what is happening, and it's particularly useful for understanding common errors.

THE SCIENCE OF SCIENCE IS EMERGING, ...
CONSIDERING THE FACTS, IT IS ...

70 pgs

Introduction to Quantum Computing and Quantum Hardware

Superconducting circuits



- 5,000+ live attendees

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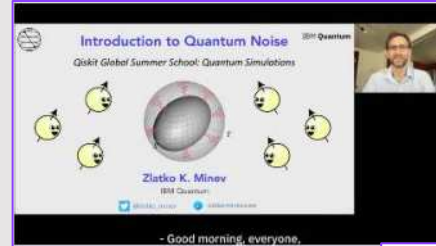
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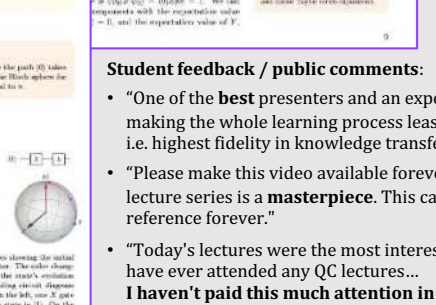
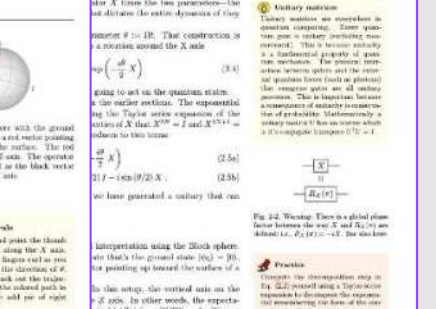
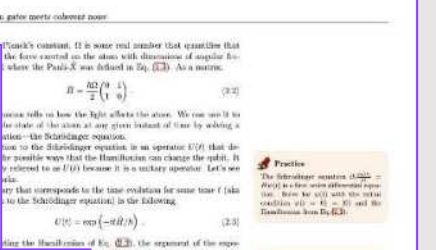
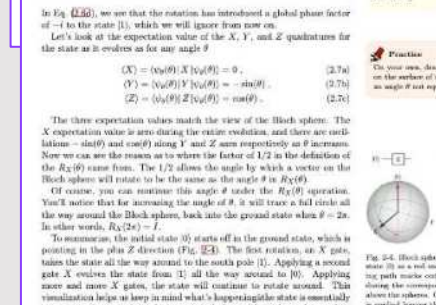
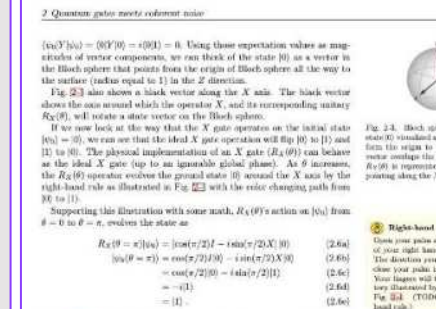
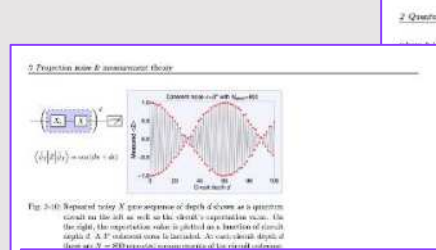
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Introduction to noise in quantum computers



- 5,000 live attendees
- 111 countries
- 27% active industry professionals
- 73% students



Introduction to quantum circuits: artificial atoms and microwave photons

Book WIP: Zlatko, Jens Koch, Michel Devoret

1. Your first qubit: from a classical circuit to a transmon (a rapid tour)

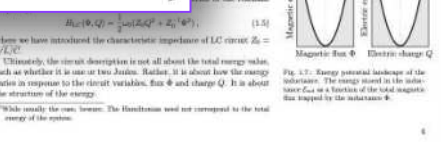
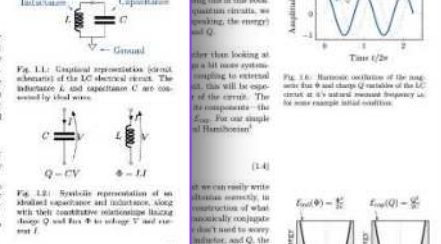
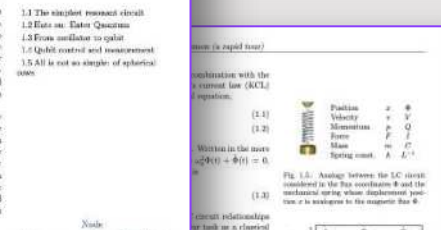
Quantum information can be stored, manipulated, and processed with the right flavor of electronic circuitry. These circuits contain increasingly more electronic charges, yet we can tune them to behave collectively — to act as if they were just one quantum degree of freedom. Engineering the properties of these degrees, we can make electrical circuits act as if they were individual atoms — artificial atoms. These form the fabric of quantum information processing. Periodically, the laws governing artificial circuits can be expressed by a relatively simple mechanical formalism that students can easily learn.

How do we describe the behavior of an artificial atom? How do we engineer them to do useful things? In this chapter, we begin our journey to find answers. We won't go deep yet, but we will go far — taking a brief large tour. We will meet the characters and variables used to specify circuit elements, and will then bridge its description to a quantum one. By enhancing this resonant circuit with non-linearity, we will make our first artificial atom and fully programmable qubit. Details and formalism will be relegated to the rest of the book. For now, have fun on your first foray into the world of quantum circuits.

1.1. The simplest resonant circuit

The LC resonant circuit. Imagine the simplest lumped circuit that resonates using wires as illustrated in Fig. 1.1. An ideal wire connects two nodes and conducts charge without any resistance or other dynamics. By selecting the circuit node associated with the lower wire as the zero voltage reference of the circuit, the ground node, we eliminate from our description one of the two nodes. The resulting circuit (see Fig. 1.1) is the simplest resonant circuit — comprising one element, one node, one inductive, and one capacitive element.

Reactive lumped elements. The LC circuit comprises lumped circuit elements that neither supply nor dissipate energy — they can only store it. Inductors and capacitors are the only elements that can store energy. As such, the two connected nodes become a single node. While in practice, the inductor L and capacitor C have a small but non-zero resistance, the ideal circuit is lossless. As we will later see, making our circuit the ground impedance from the description of the circuit is a subtle degree of freedom.



Student feedback / public comments:

- "One of the best presenters and an expert in making the whole learning process least noisy; i.e. highest fidelity in knowledge transferring."
- "Please make this video available forever. This lecture series is a masterpiece. This can be our reference forever."
- "Today's lectures were the most interesting I have ever attended any QC lectures... I haven't paid this much attention in long time and Zlatko made a huge impact, Thank you."

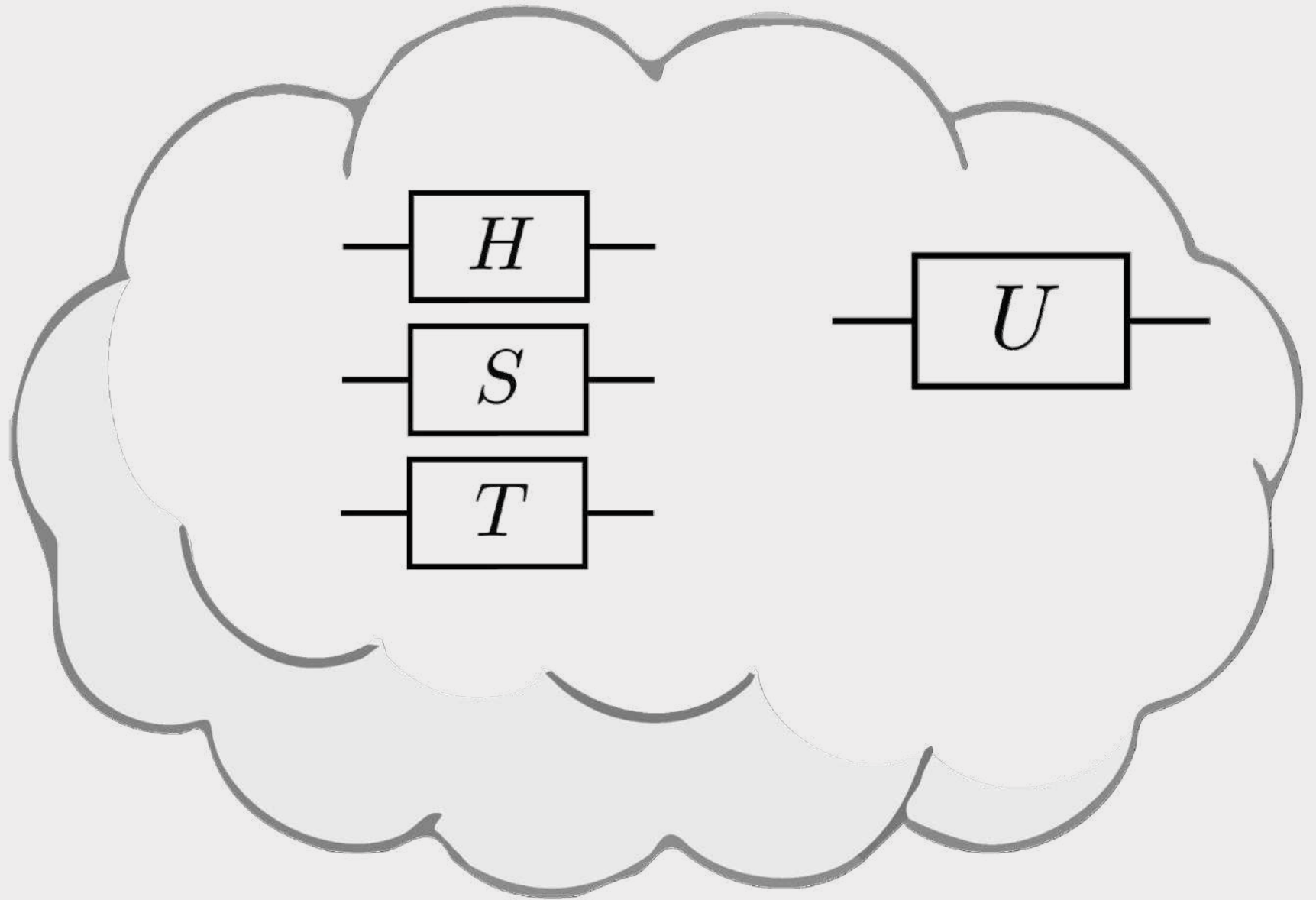


Open Labs

Est. 2012

Quantum Computer and IBM Quantum

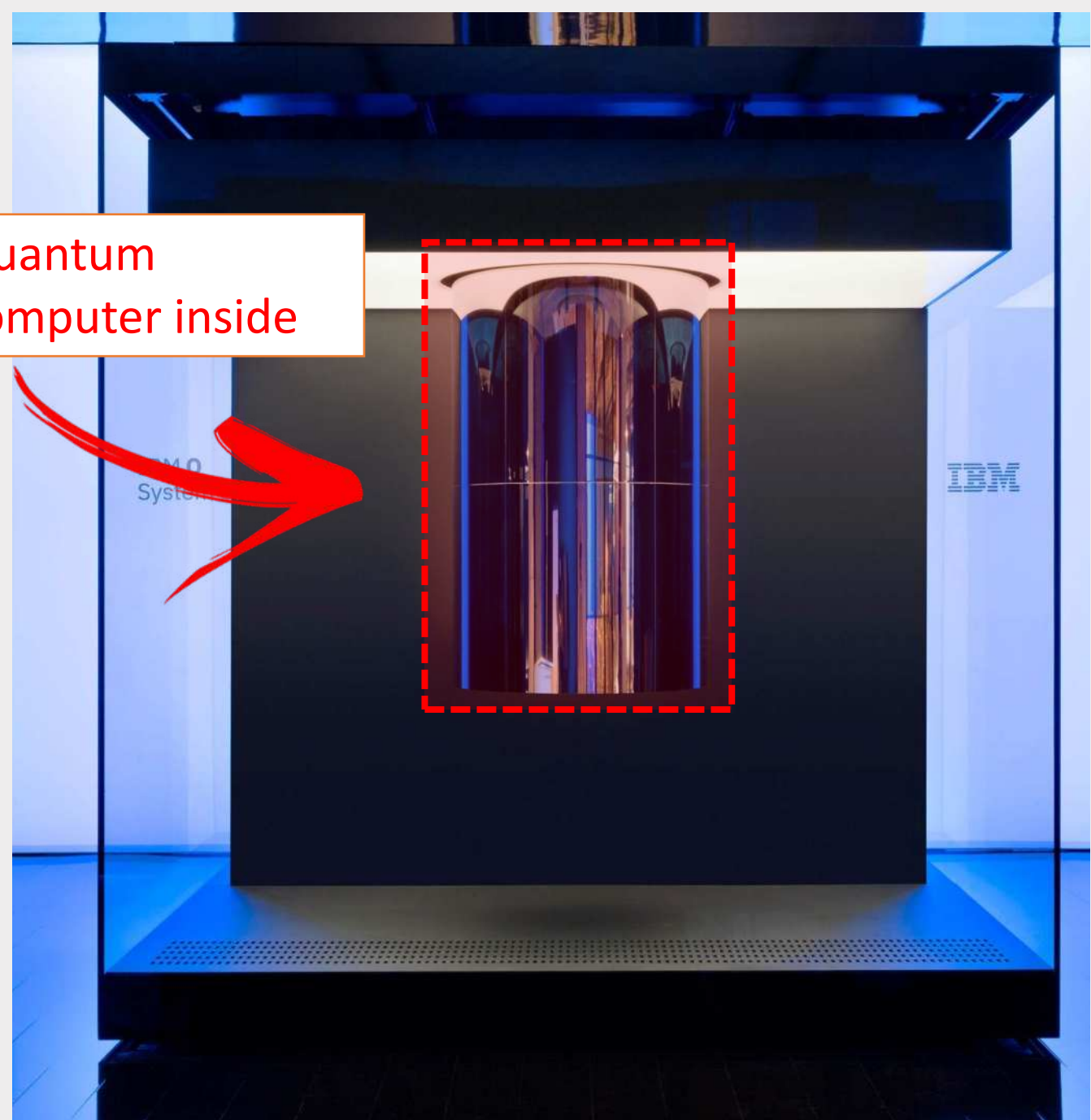
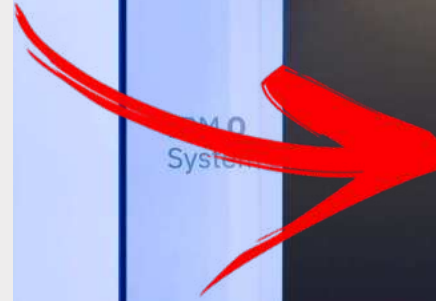
"Quantum cloud"



Quantum computer

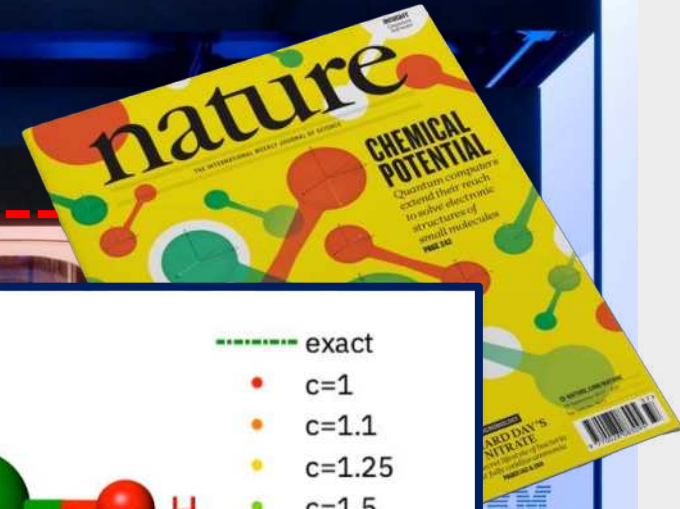


Quantum
computer inside



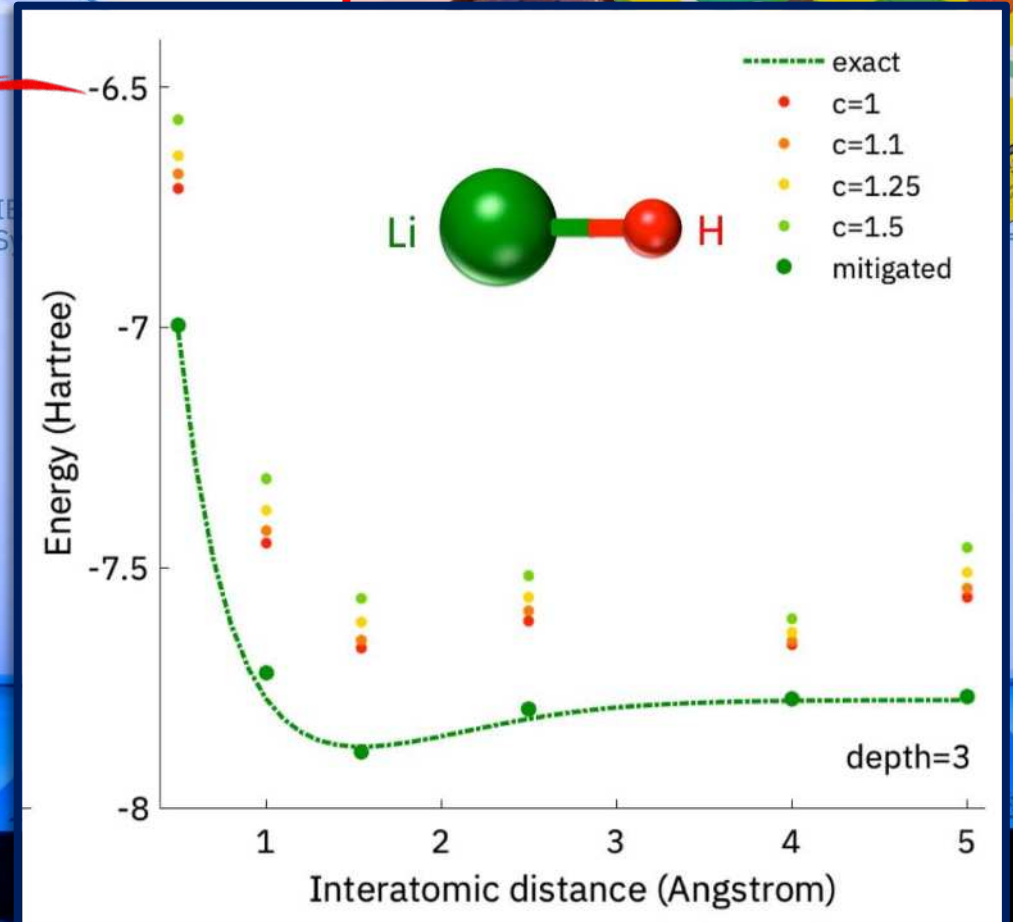
Programming quantum

execute program
in Python



```
(Qiskit) → qprogram_examples git:(master) x
```

I



Biggest challenge?

Please do share

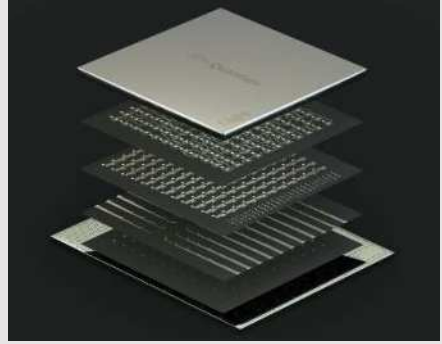


Biggest challenge

Noise
(Errors)

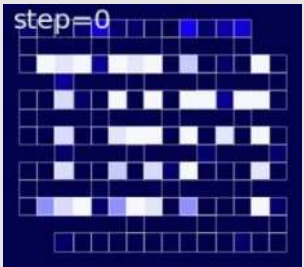


Quantum computational science on quantum hardware

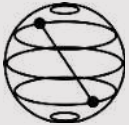


What should I know about using today's quantum hardware?

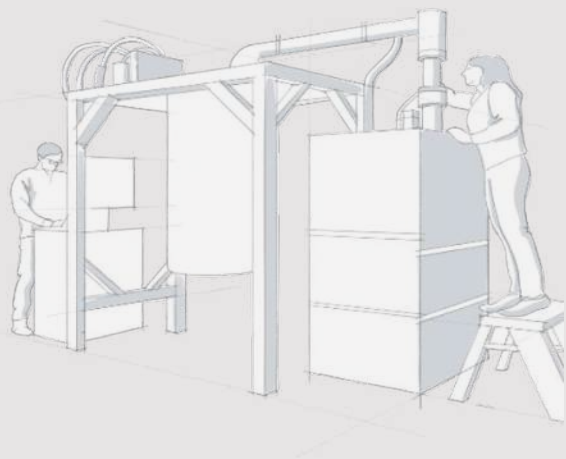
Quantum and Noise Introduction in a nutshell



1 qubit (Hello World!)
to
100+ qubits (state of art)



Hello World: Is it even or odd?



Compute resources

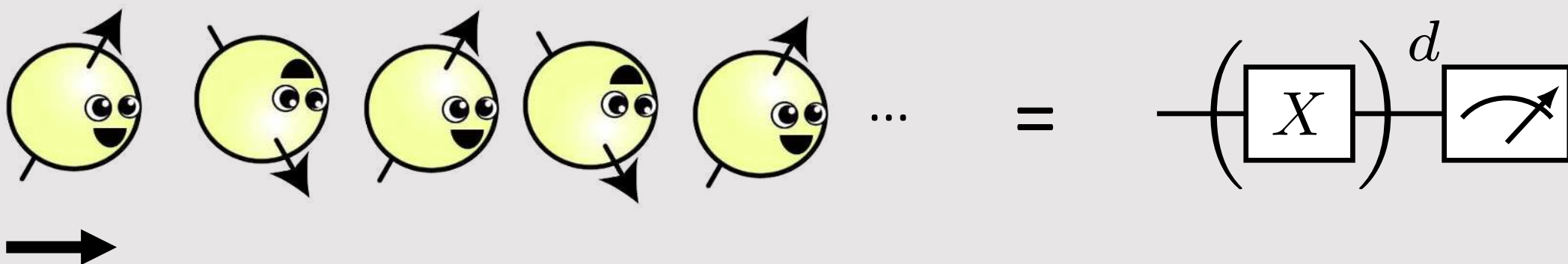
Algorithm input

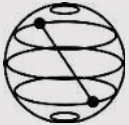
Task

1 qubit 
an integer $d > 0$
even or odd?

A toy algorithm

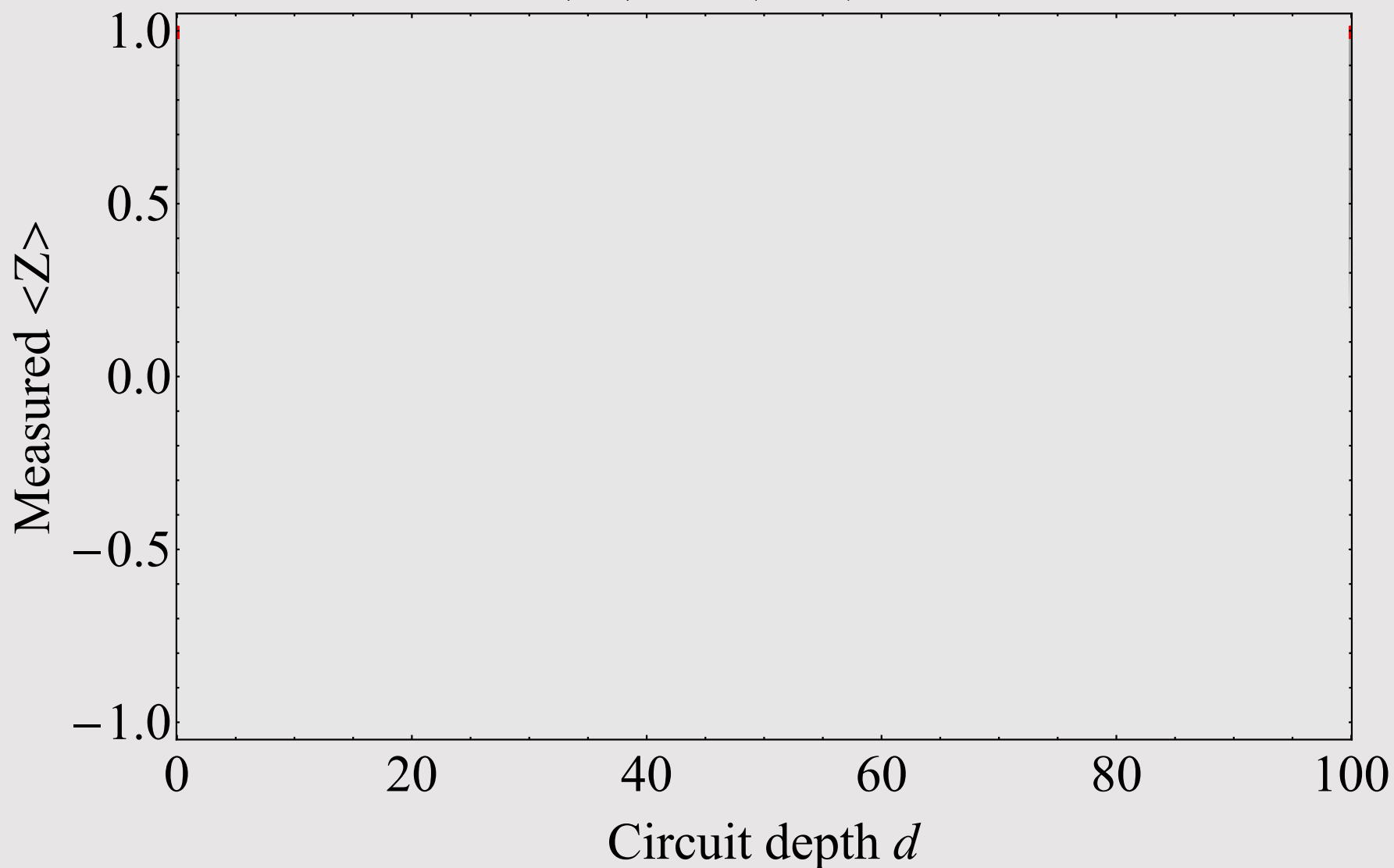
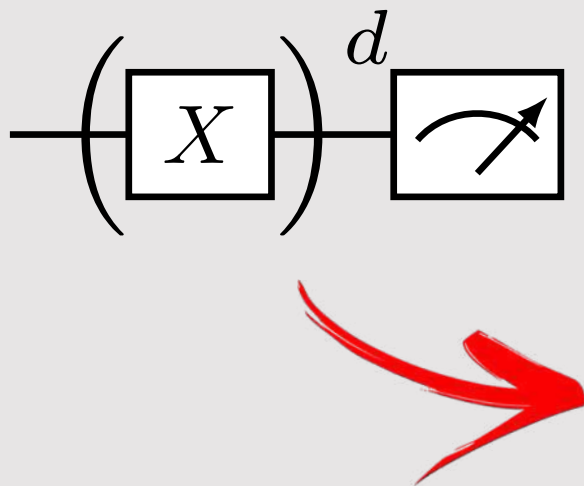
Start in 0, flip d times, measure

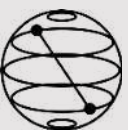




Hello World! Ideal expectation results

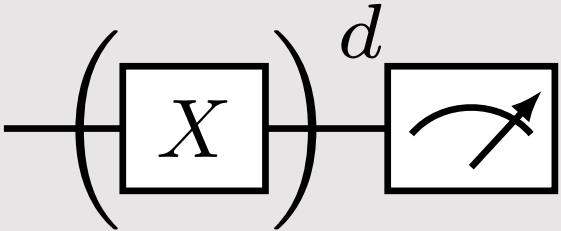
$$\langle Z \rangle = (-1)^d$$



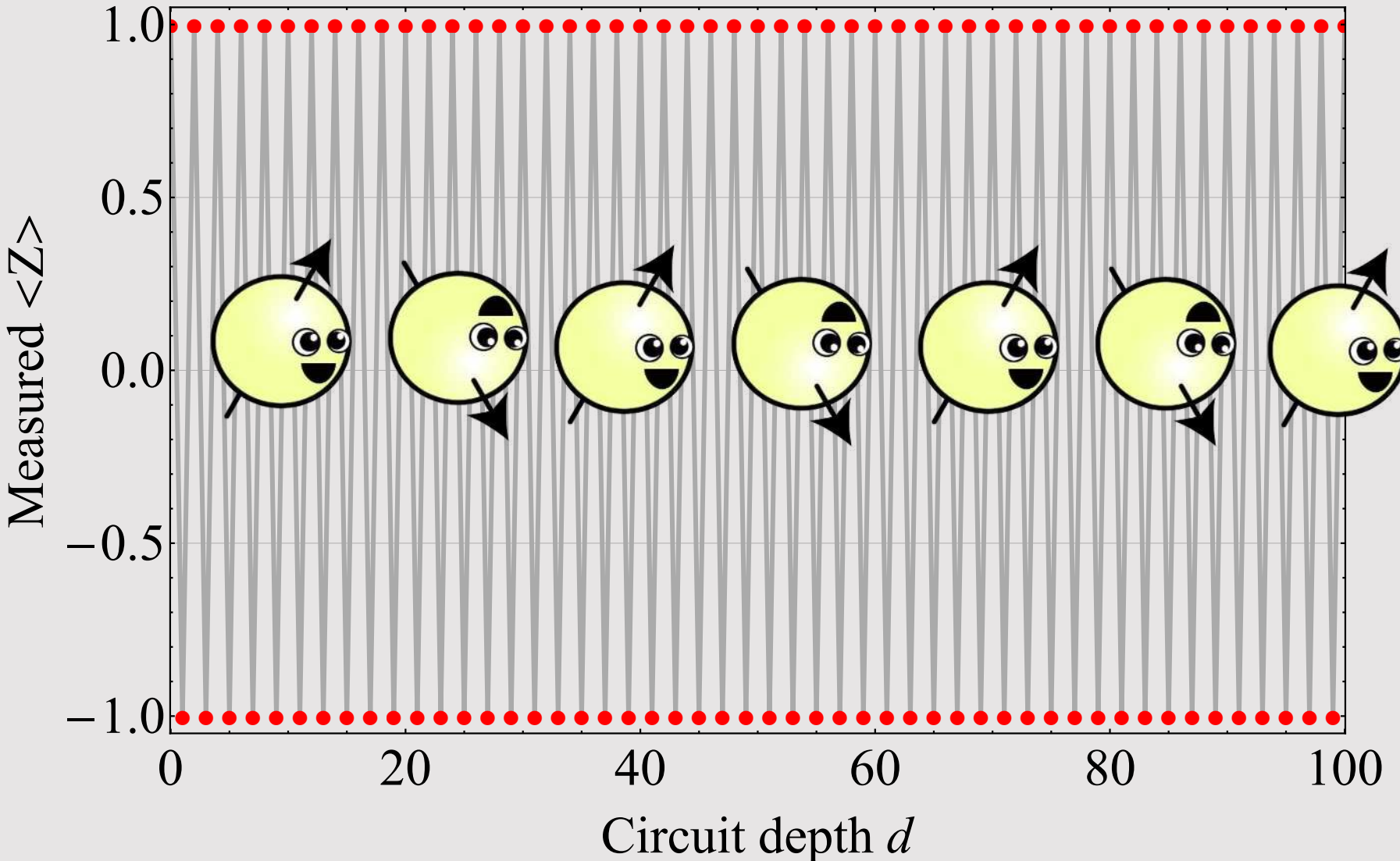
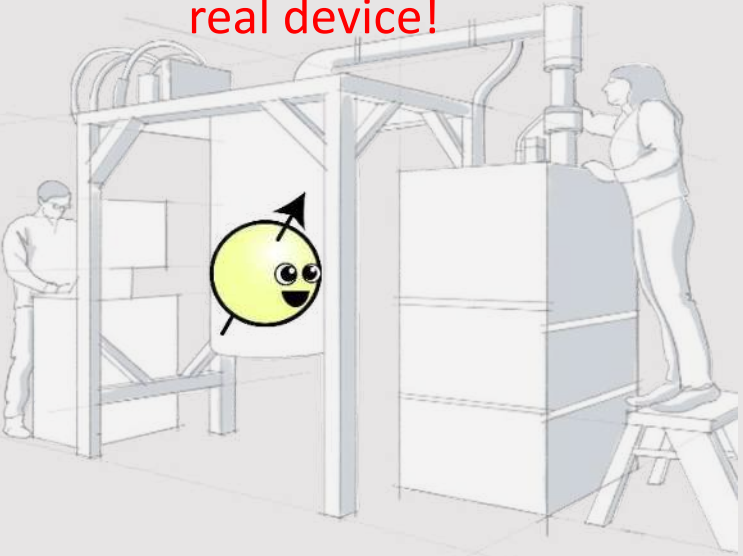


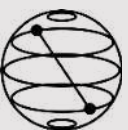
Hello World! Ideal expectation results

$$\langle Z \rangle = (-1)^d$$

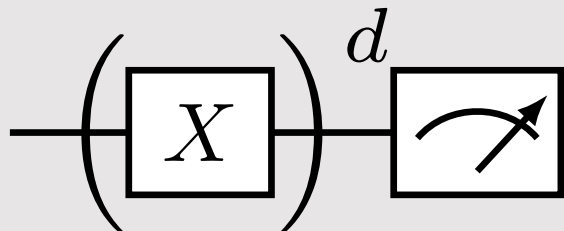


Let's run on an (old) real device!

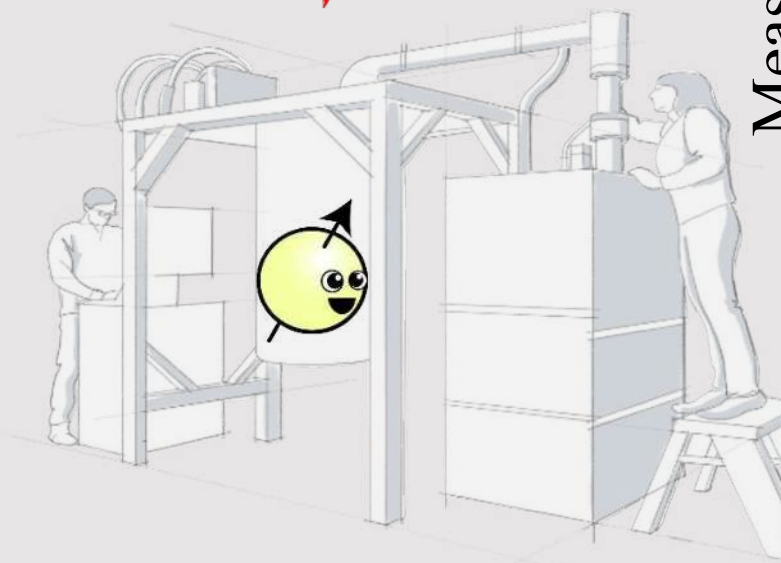




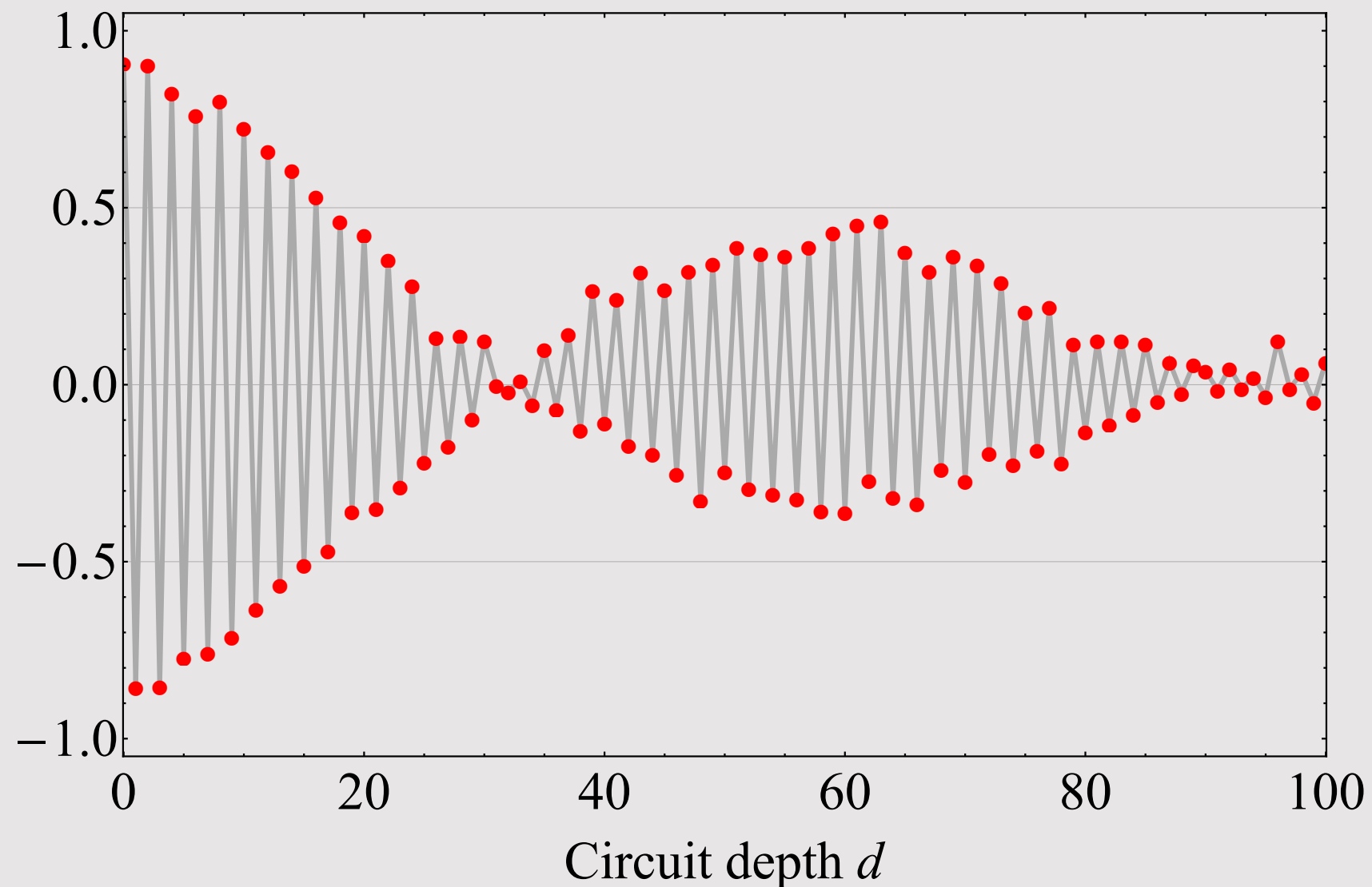
Hello World! Real expectation results

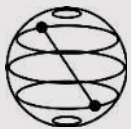


Raw data!?!?



Measured $\langle Z \rangle$



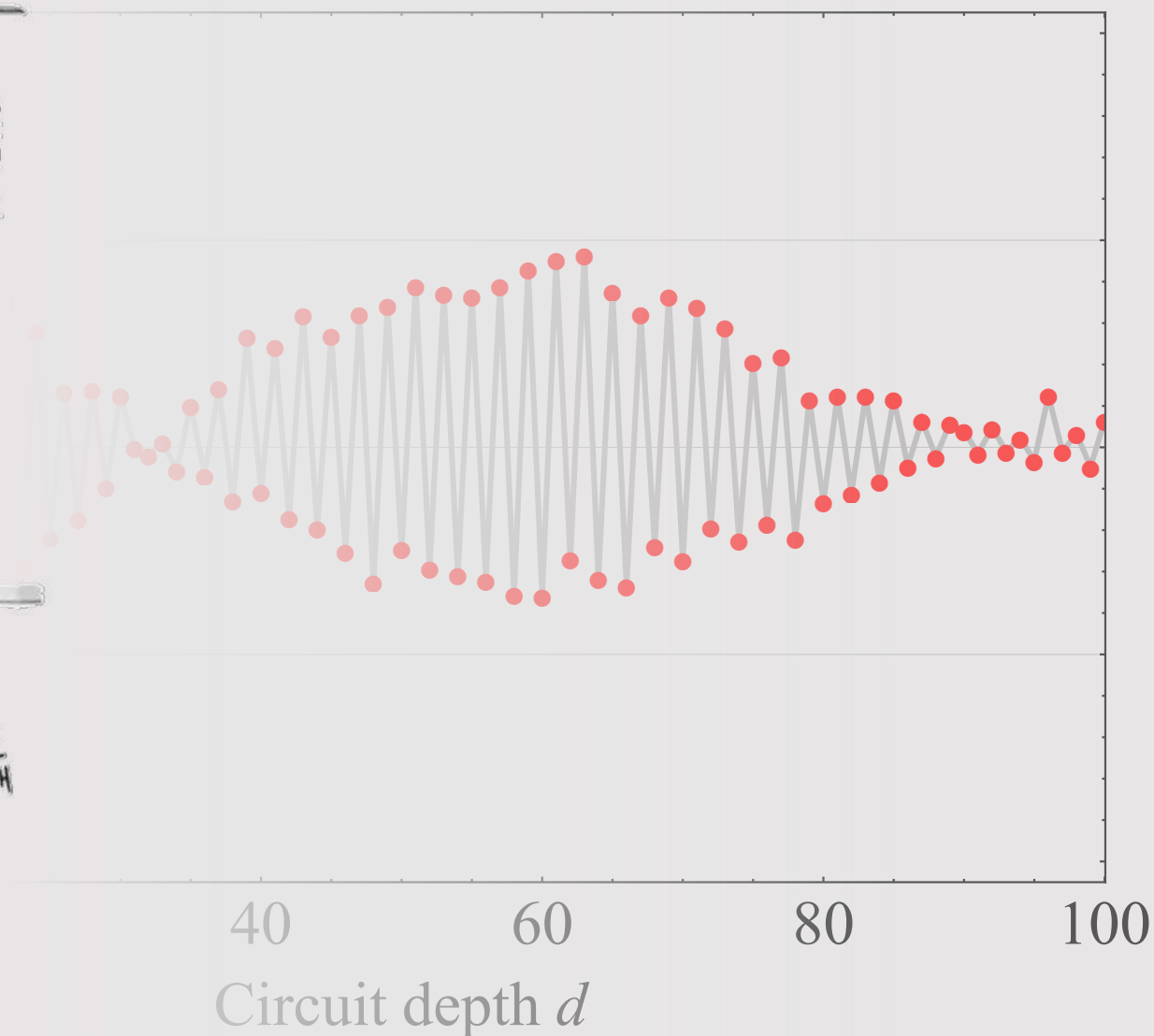


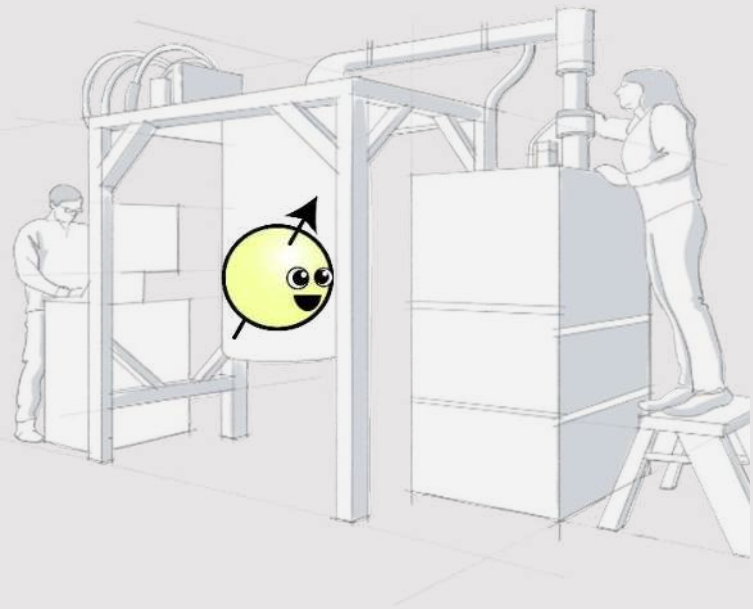
Real & noisy quantum processors: Why study noise?



“Well, your quantum computer is broken in every way possible simultaneously.”

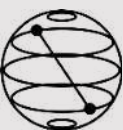
PAUL
NOTH



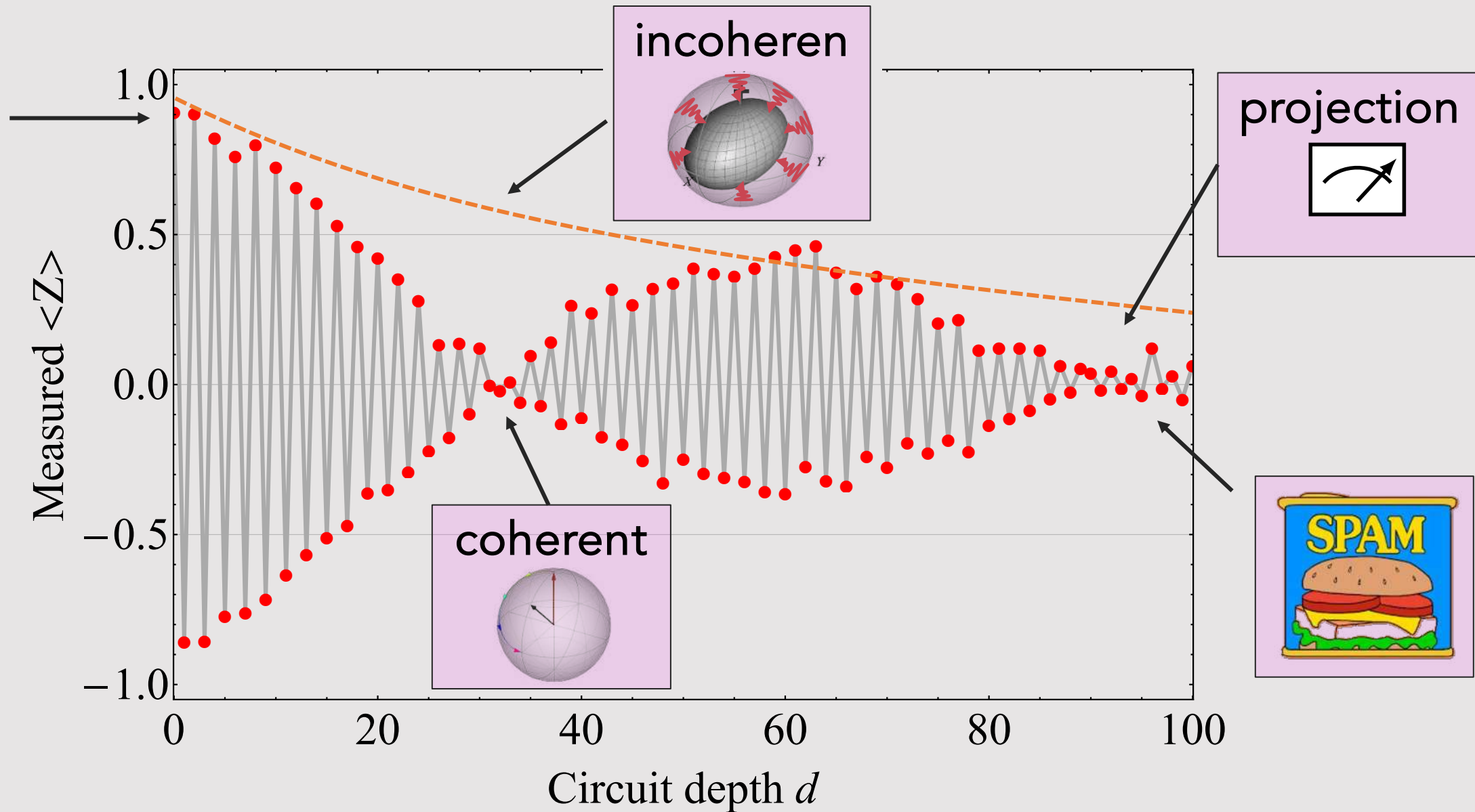


"Quantum phenomena
do *not* occur in a Hilbert space,
they occur in a laboratory."

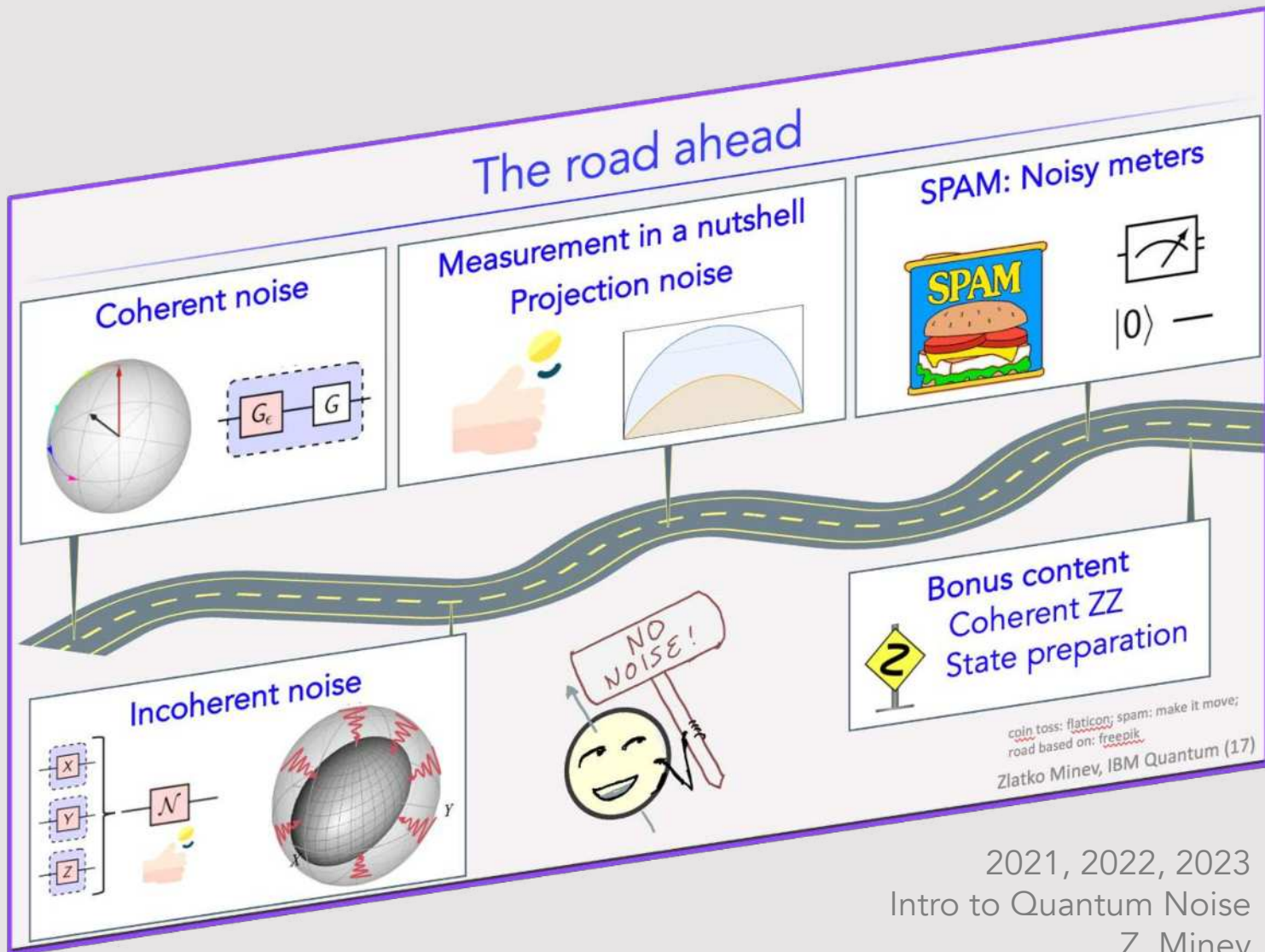
Asher Peres



Elements of 🤯 noise



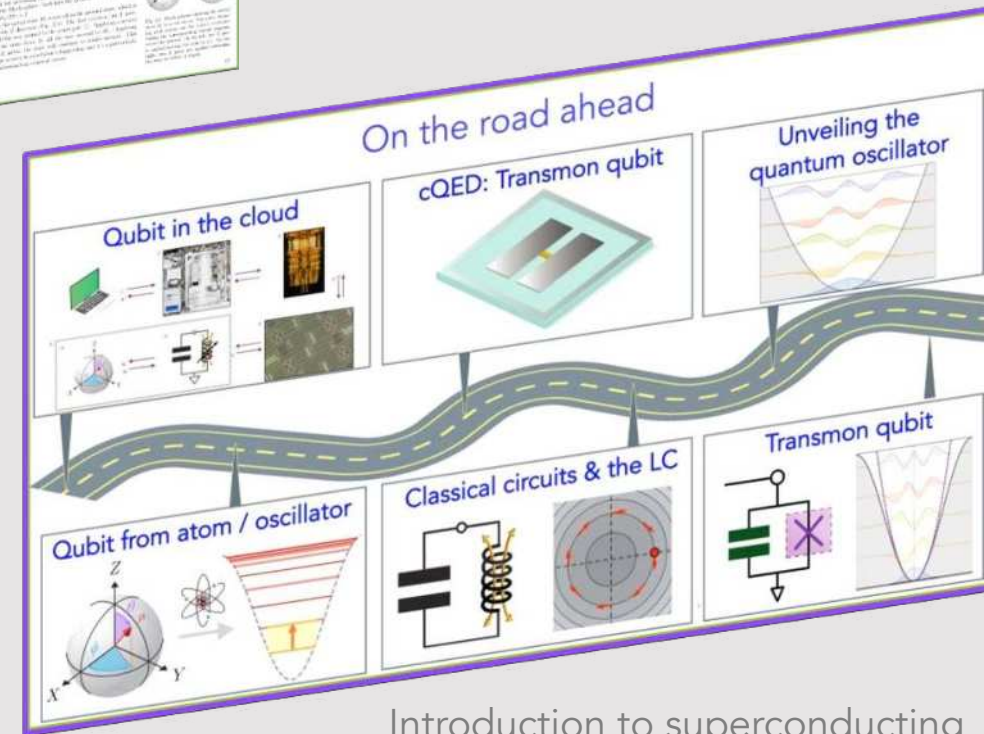
Learn the basics?



2021, 2022, 2023
Intro to Quantum Noise
Z. Minev

qiskit.org/learn

Qiskit Global Summer School 2023 coming up soon!
5,000+ students



How to deal with errors due to noise?

Monitor
Error occurs
Error detect



Quantum error correction

Shor, PRA (1995), ...

Monitor
Error anticipated
Tell signal detected



Catch and reverse

Minev, Nature (2019), ...

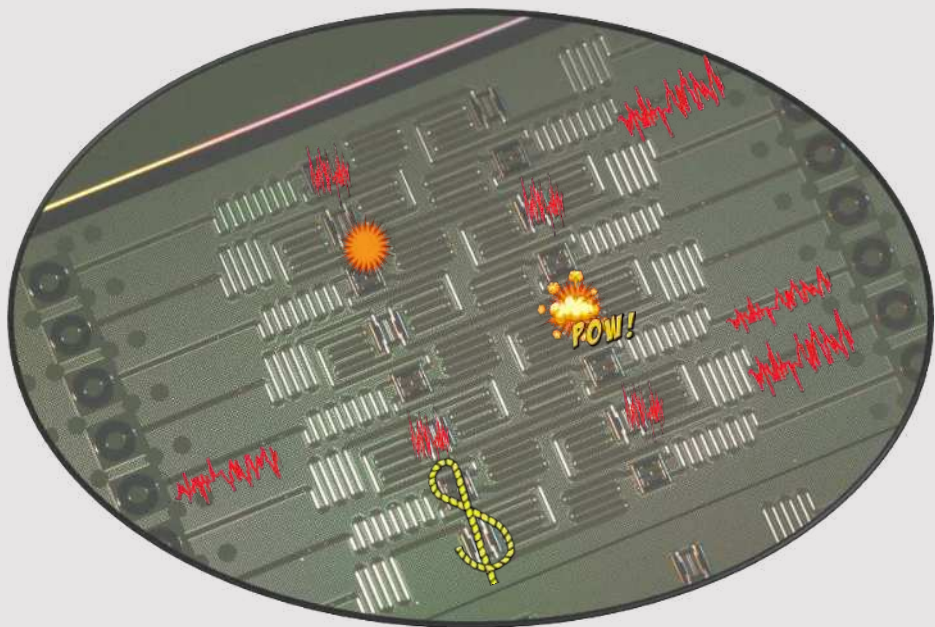
No monitor
Error occurs
Error undetected



Error mitigation

... subject of today

Error mitigation and error correction at a glance



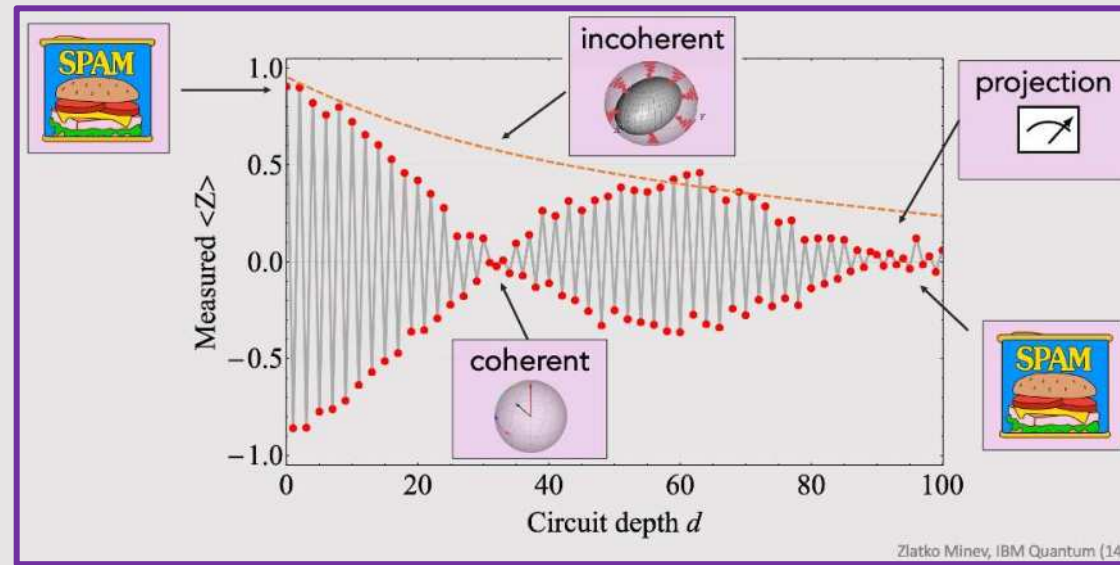
Error mitigation Working with what you have

- **benefit** suppress errors on classical results (expectation values)
- **q-cost** no extra qubits or hardware resources needed
- **c-cost** trades classical resources (post-processing/runtime) for lower error
- **limitation** bad asymptotic scaling: high number of samples & circuits

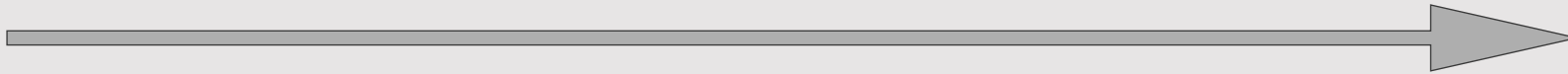
Error correction Protecting your quantum information

- **benefit** suppress & correct errors to arbitrarily small level
- **q-cost** very large qubit and hardware overhead
- **c-cost** decoding and encoding can be classically costly
- **challenge** requires fault-tolerant operations and readout

Error mitigation landscape



more
speed

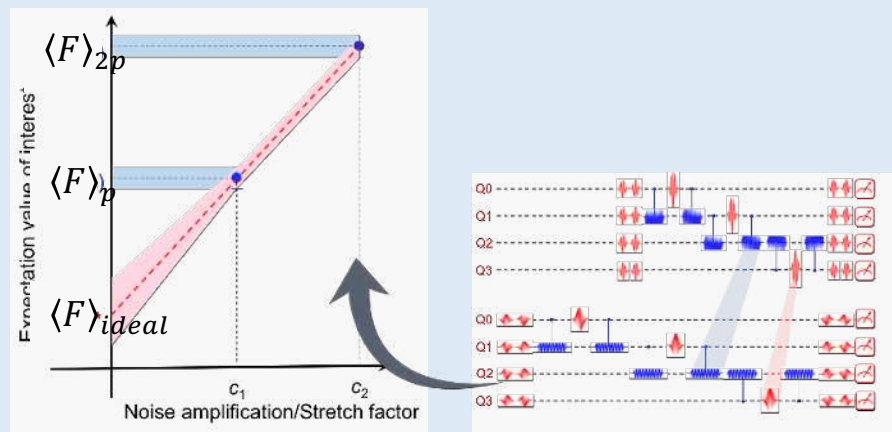


more information,
accuracy



Error mitigation landscape

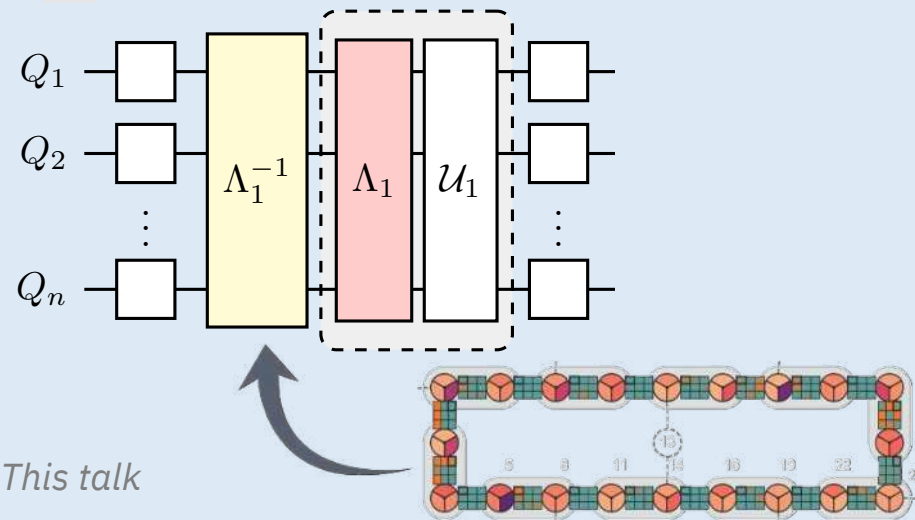
Zero-noise extrapolation (ZNE)



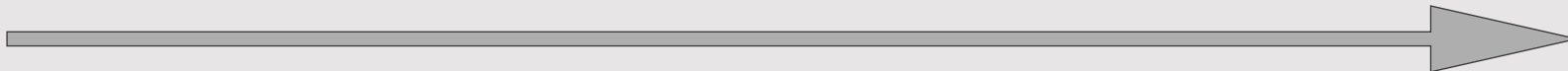
Nature 567, 491 (2019)

54

Probabilistic error cancellation (PEC)



more
speed

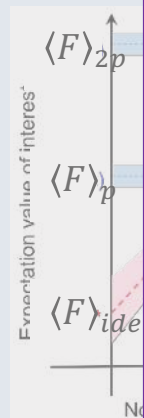


more information,
accuracy

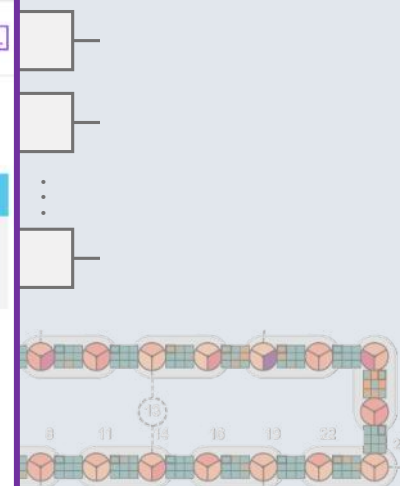


Error mitigation landscape

Zero-noise extrapolation (ZNE)



Probabilistic error cancelation (PEC)



more
speed

more information,
accuracy

 **Qiskit**

qiskit.org

Overview Learn Community

English

Search Qiskit Runtime IBM

Get started

Overview

Getting Started

backend.run vs. Qiskit Runtime

Introduction to primitives

Tutorials

Get started with Estimator

Get started with error suppression and error mitigation

VQE with Estimator

CHSH with Estimator

Get started with Sampler

Qiskit Runtime IBM Client documentation > Error suppression and error mitigation with Qiskit Runtime

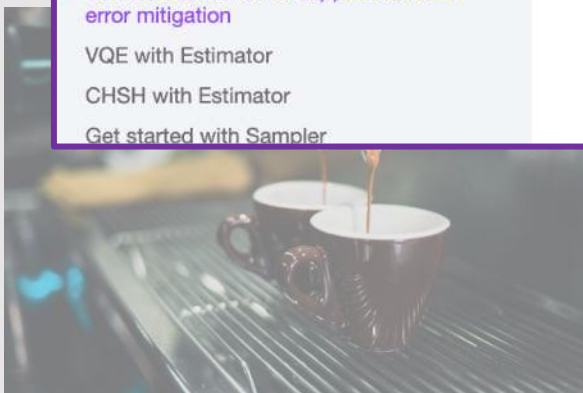
NOTE

This page was generated from docs/tutorials/Error-Suppression-and-Error-Mitigation.ipynb.

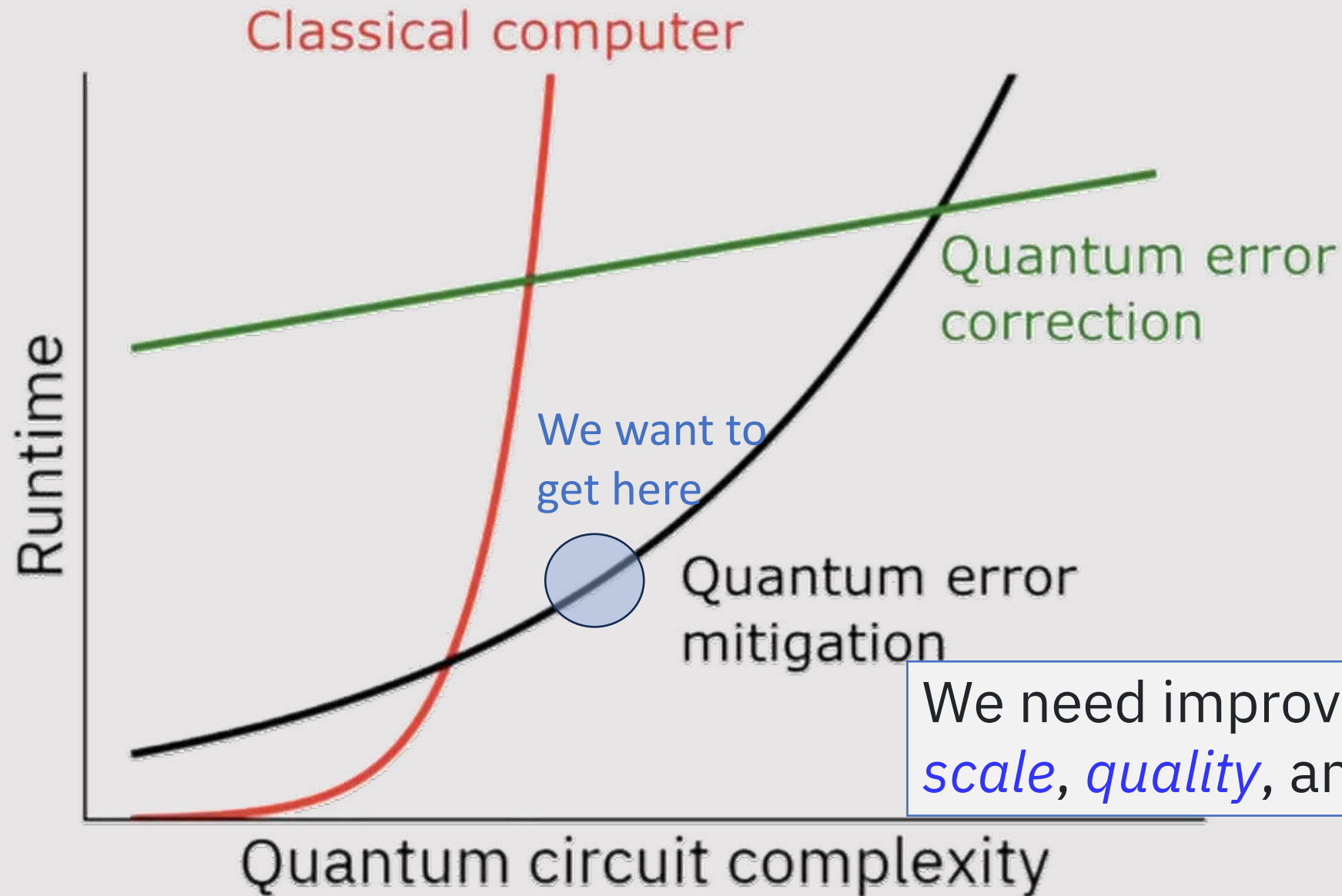
Error suppression and error mitigation with Qiskit Runtime

```
[1]: import datetime
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt
plt.rcParams.update({"text.usetex": True})
plt.rcParams["figure.figsize"] = (6,4)
mpl.rcParams["figure.dpi"] = 200

from qiskit_ibm_runtime import Estimator, Session, QiskitRuntimeService,
Options
from qiskit.quantum_info import SparsePauliOp
from qiskit import QuantumCircuit
```



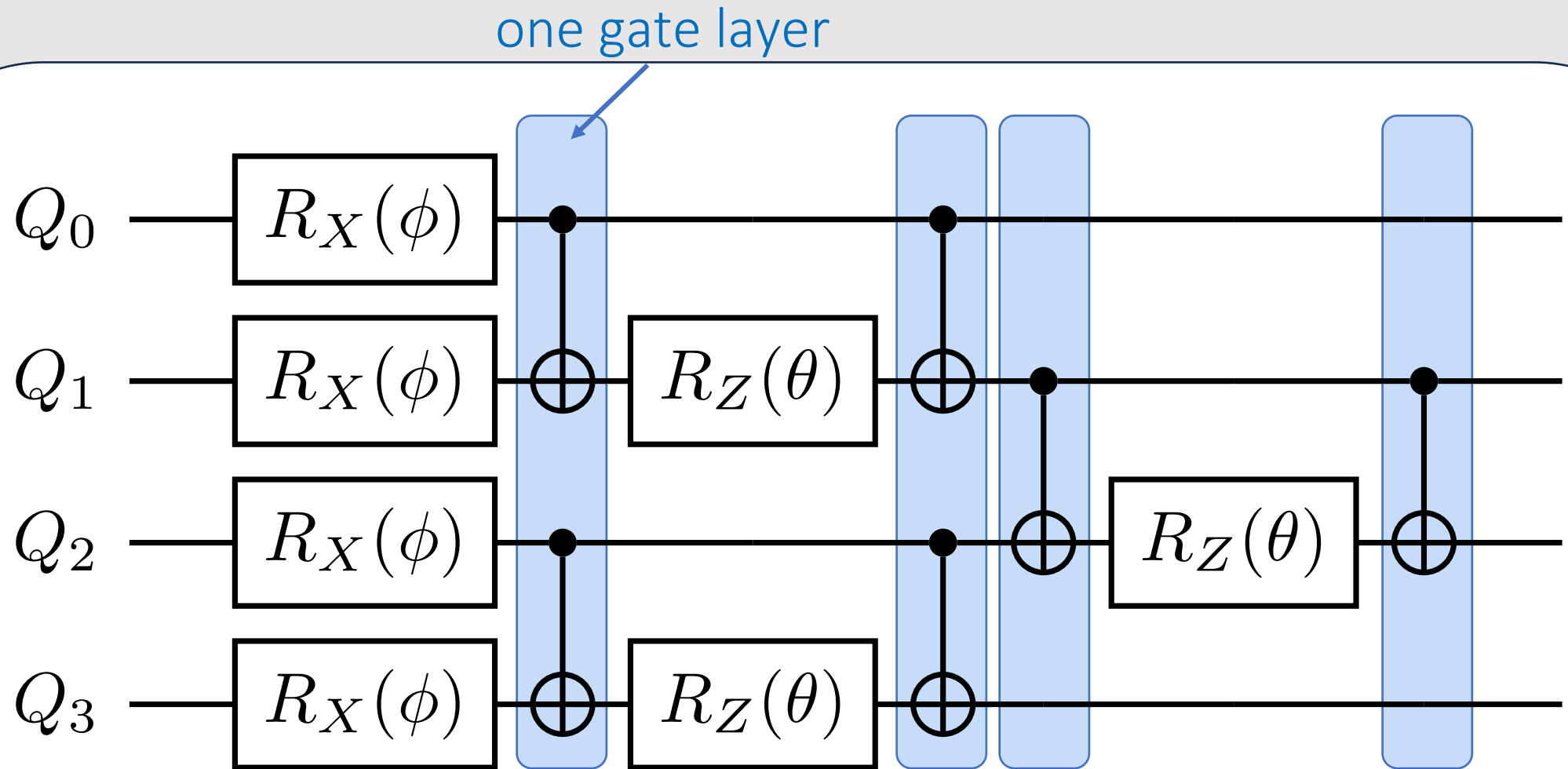
Path to utility of quantum computers before error correction



We need improvements in *scale*, *quality*, and *speed*.

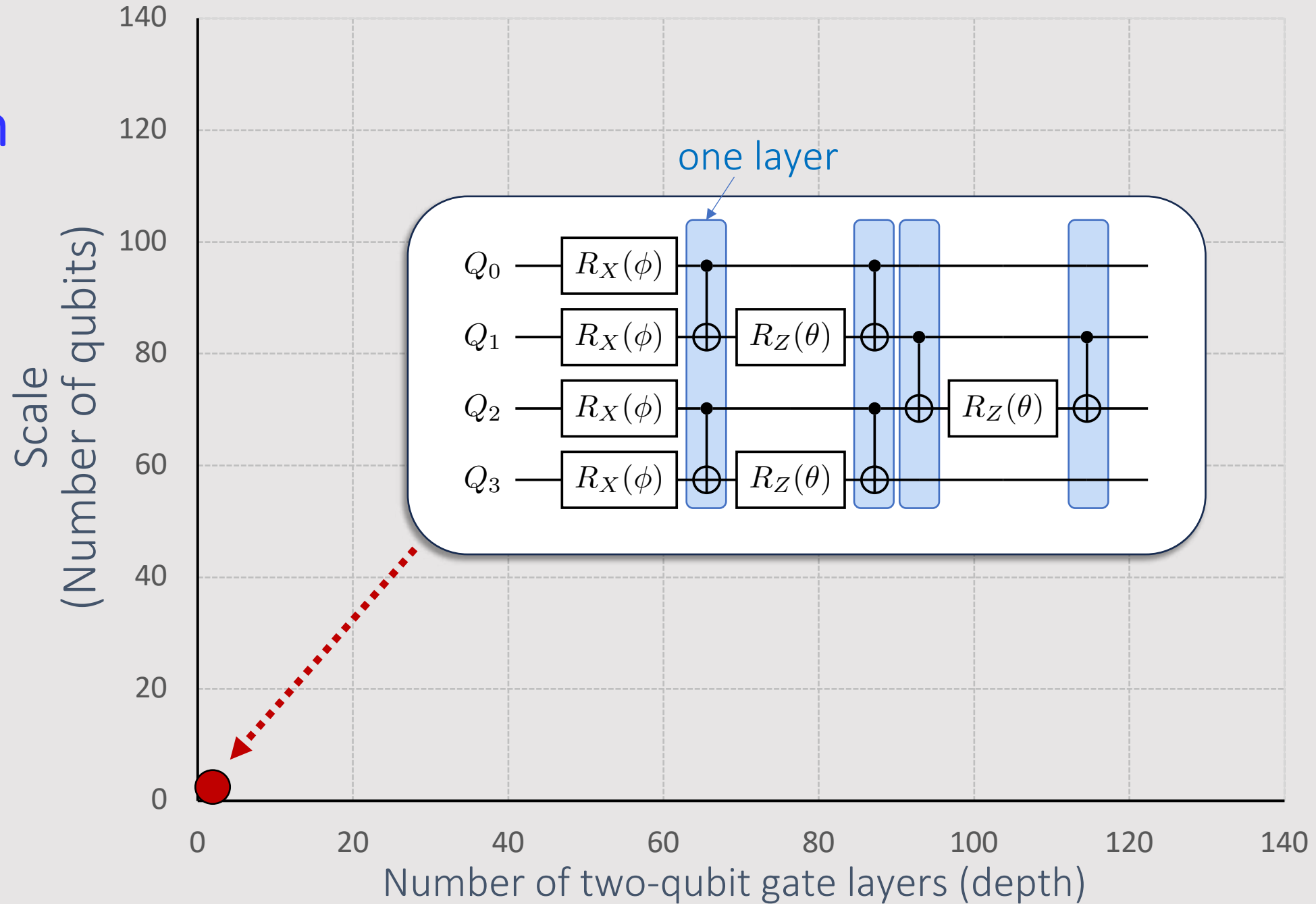
Bring useful quantum computing

A prequel to the panel



4 qubits x 4 gate layers

Circuit
width $\times$ depth



Many-body physics on superconducting quantum computers

A representative subset of
many-body physics
experiments on
superconducting quantum
computers until 2023,
depths approximate*

[1] Science 369, 1084 (2021)

[2] Nature 594, 508 (2021)

[3] Science 374, 1479 (2021)

[4] Nature 601, 531 (2022)

[5] Science 378, 785 (2022)

[6] Nature 612, 240 (2022)

[7] Nature 618, 264 (2023)

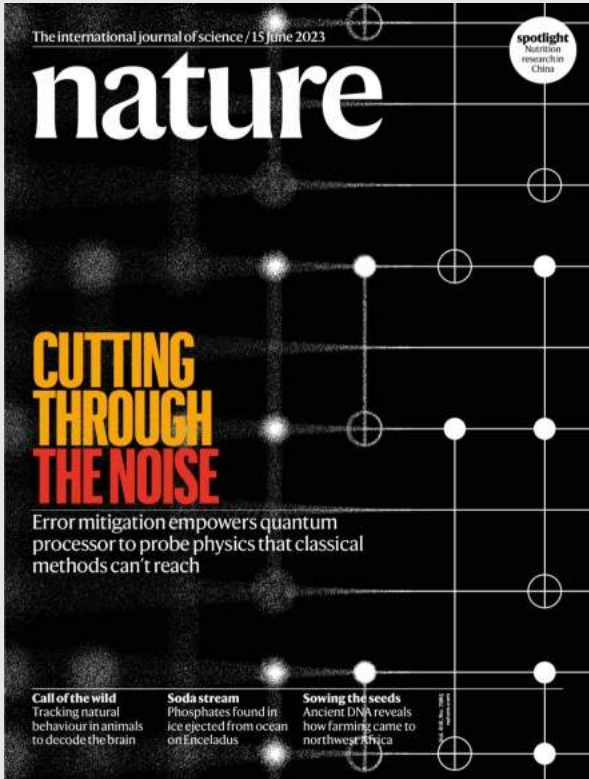
...

* signal stops or decays more than
50% beyond this depth, capped at 140

* note: quantum advantage with shallow
circuits, Bravyi, Gosset, Konig, and related



Some experiments in last 3 months

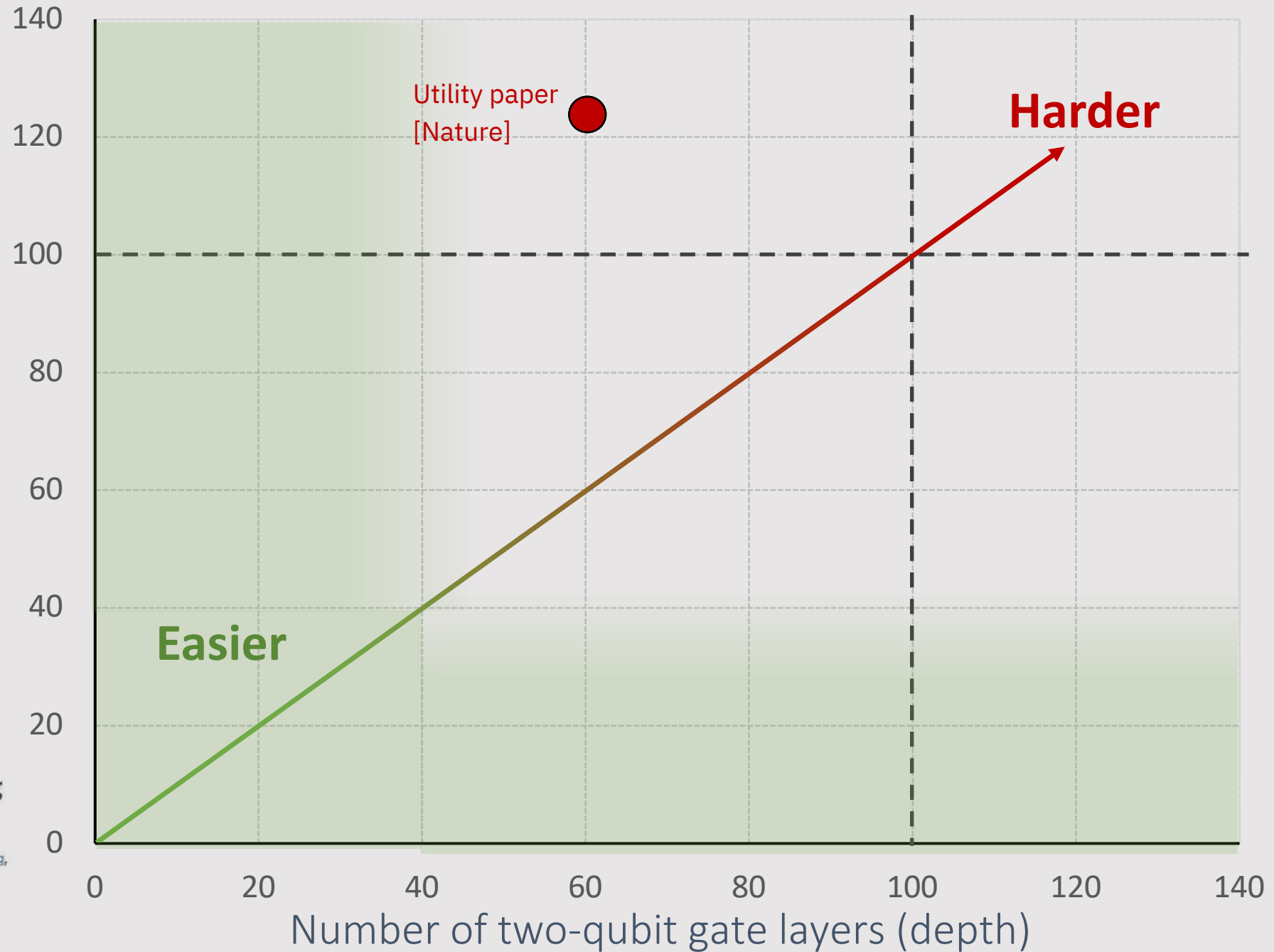


Evidence for the utility of quantum computing before fault tolerance

[Youngseok Kim](#), [Andrew Eddins](#), [Sajant Anand](#), [Ken Xuan Wei](#), [Ewout van den Berg](#),
[Sami Rosenblatt](#), [Hasan Nayfeh](#), [Yantao Wu](#), [Michael Zaletel](#), [Kristan Temme](#) & [Abhinav
Kandala](#)

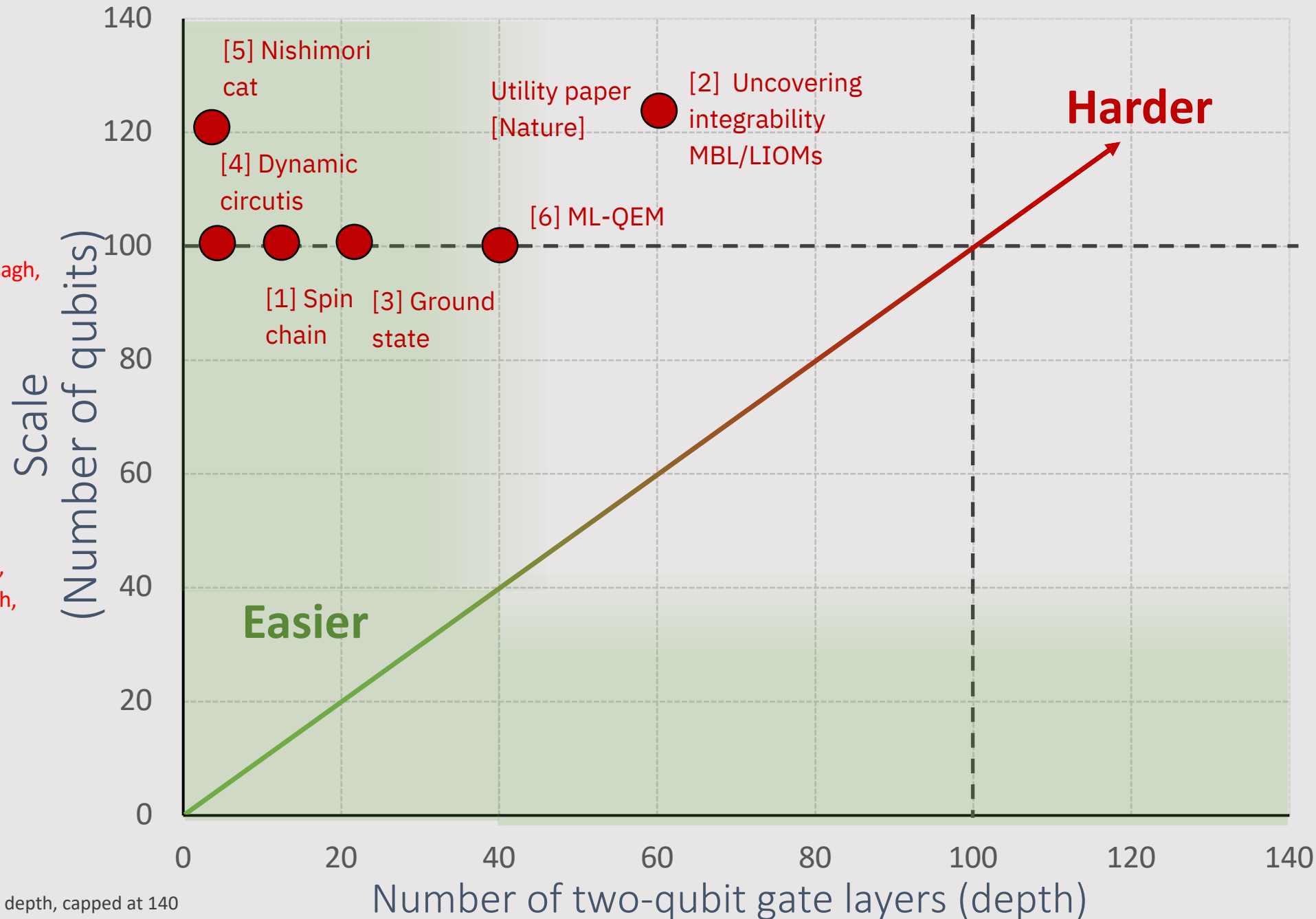
[Nature](#) 618, 500–505 (2023) | [Cite this article](#)

Scale
(Number of qubits)



Some recent experiments

- [1] Yu, Zhao, Wei.
arXiv: 2207.09994 (2022)
- [2] Shtanko, Wang, Zhang, Harle, Seif, Movassagh, Minev.
arXiv: 2307.07552 (2023)
- [3] Farrell, Illa, Ciavarella, Savage.
arXiv: 2308.04481 (2023)
- [4] Bäumer, Tripathi, Wang, Rall, Chen, Majumder, Seif, Minev.
arXiv: 2308.13065 (2023)
- [5] Chen, Zhu, Verresen, Seif, Bäumer, Layden, Tantivasadakarn, Zhu, Sheldon, Vishwanath, Trebst, Kandala.
arXiv: 2309.02863 (2023)
- [6] Liao, Wang, Sitdikov, Salcedo, Seif, Minev.
arXiv: 2308.13065 (2023)
- ...



* signal stops or decays more than 50% beyond this depth, capped at 140

* note: quantum advantage with shallow circuits, Bravyi, Gosset, Konig, and related



2019
Falcon
27 Qubits



2020
Hummingbird
65 Qubits



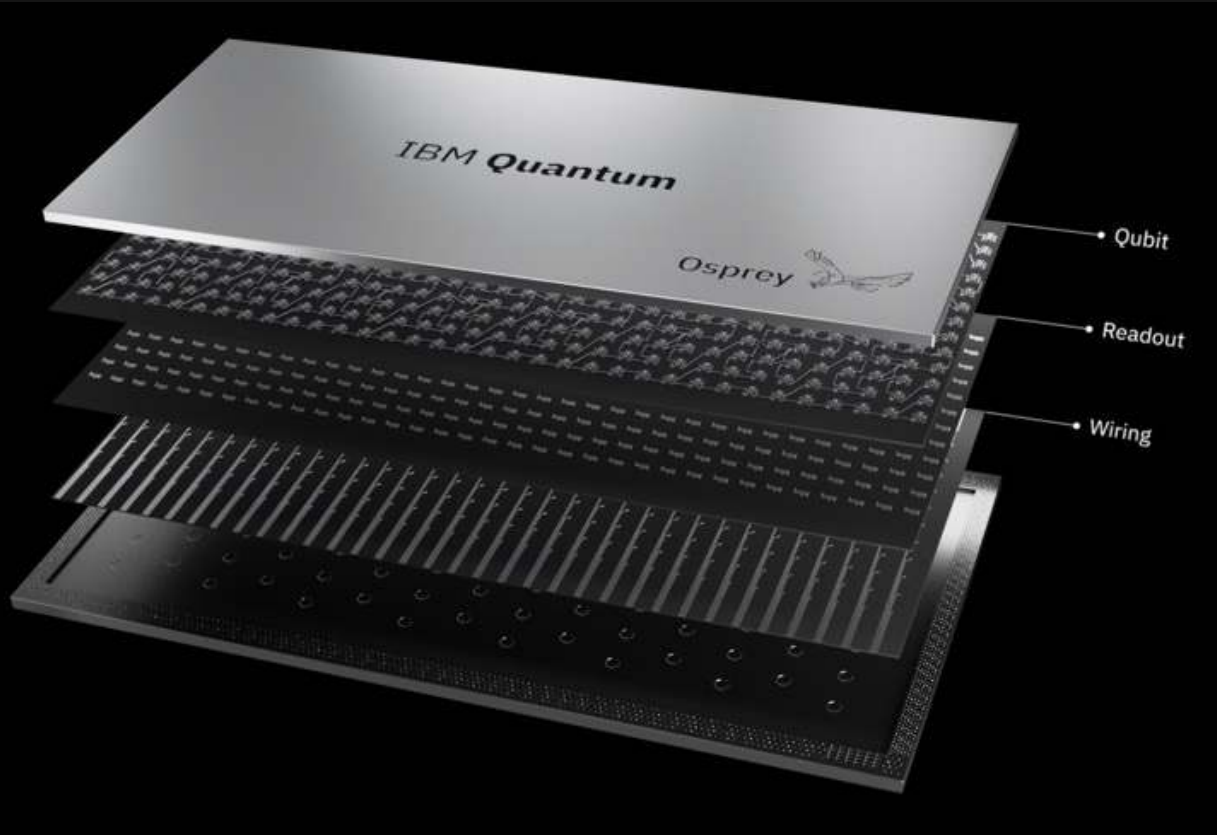
2021
Eagle
127 Qubits



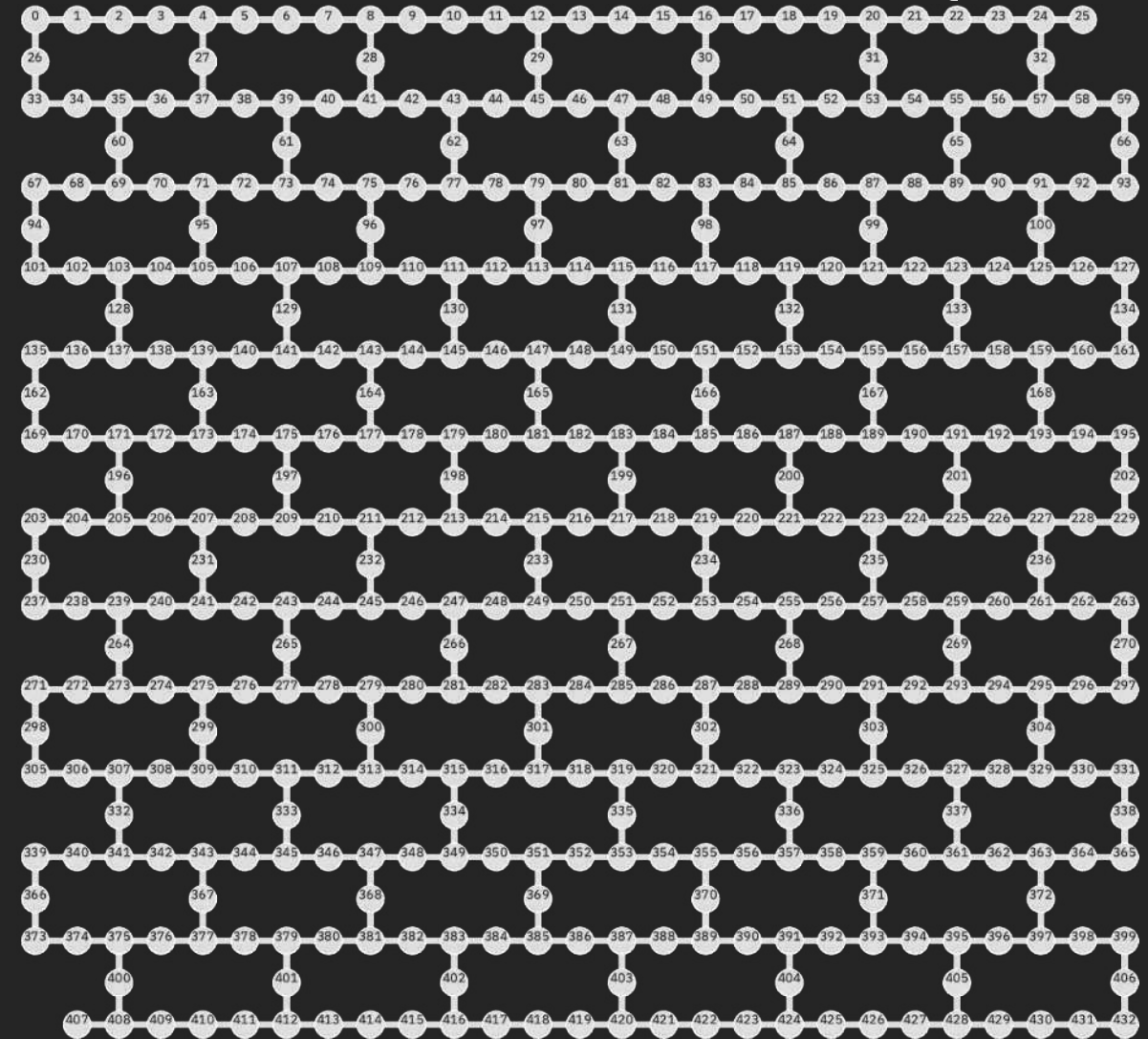
2022
Osprey
433 Qubits



Osprey – 433 Qubits



IBM Quantum



Osprey connectivity map

Performance =

Scale

Number of qubits

+ Quality

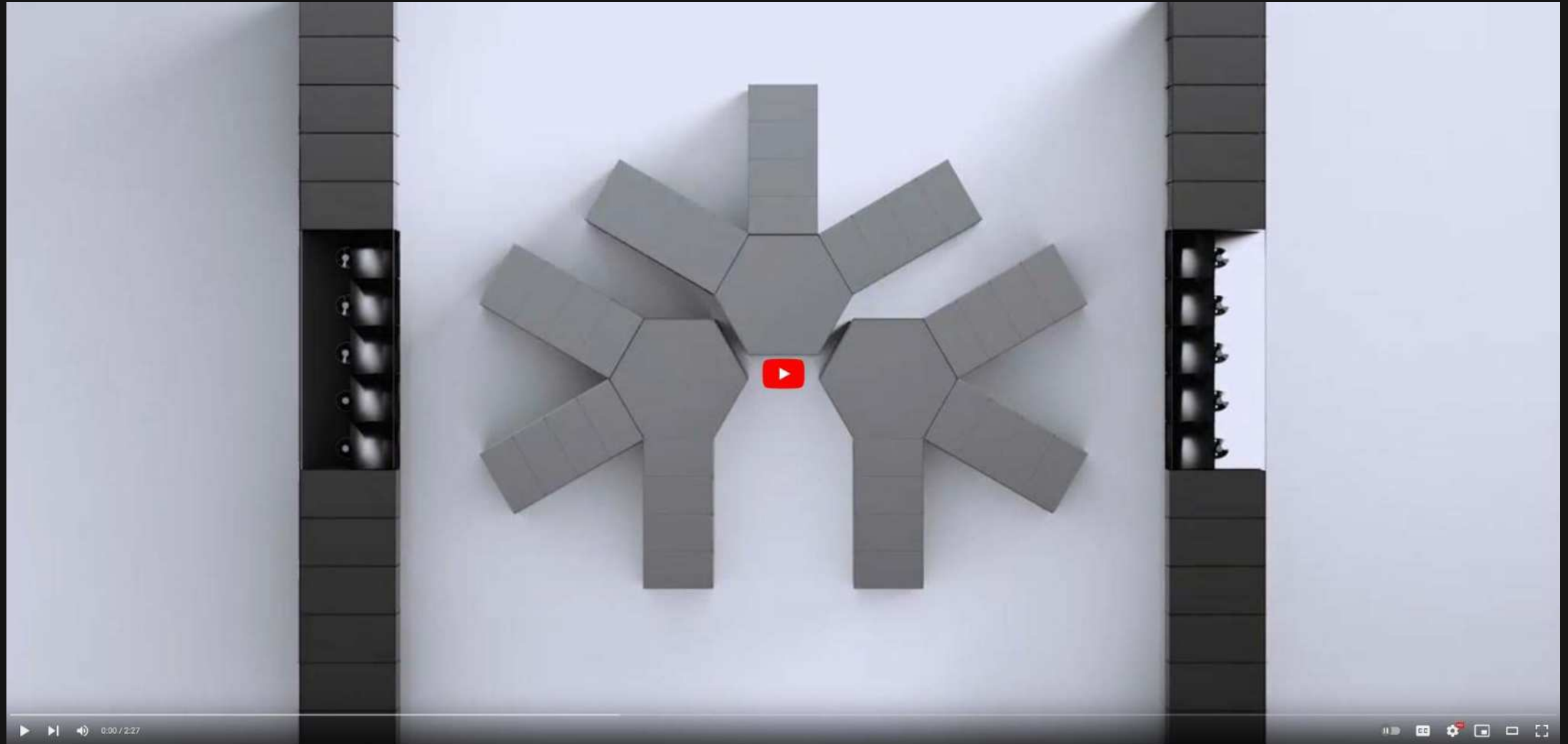
Circuit fidelity

+ Speed

Circuit execution speed

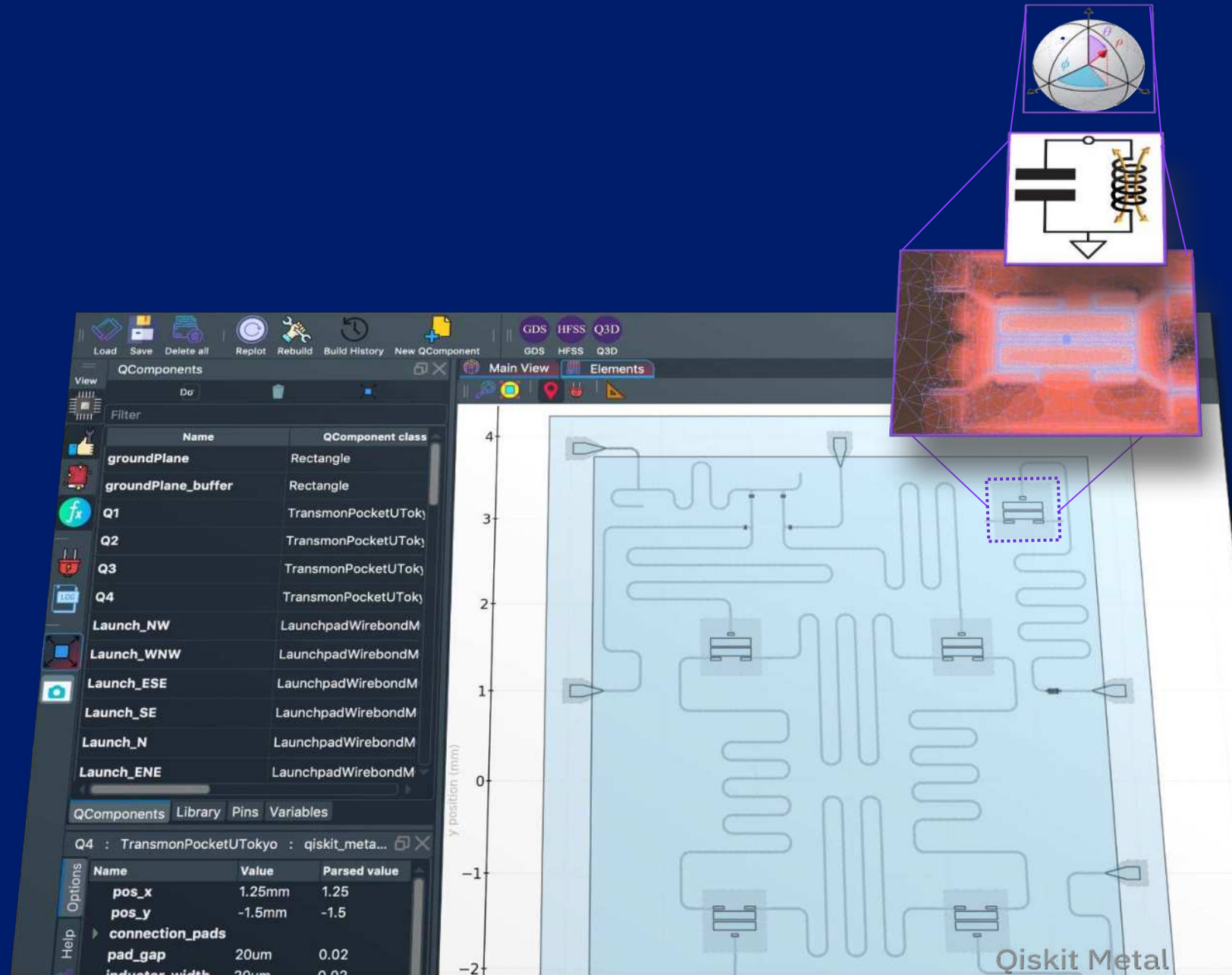
100K-qubit Quantum-Centric Supercomputer

IBM Quantum



https://www.youtube.com/watch?v=7aa_ik_UYTw

Qiskit Metal



Technology stack

Gates & time-
evolution analysis

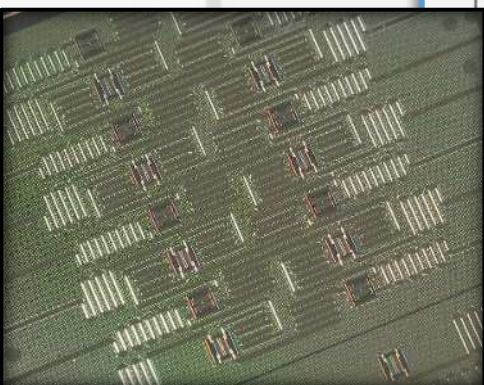
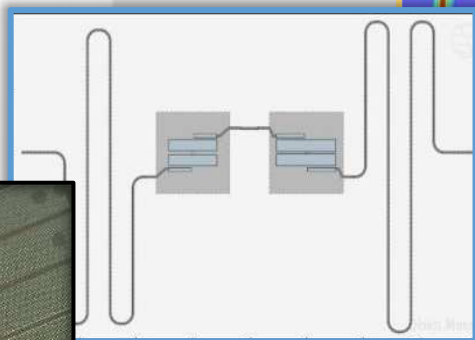
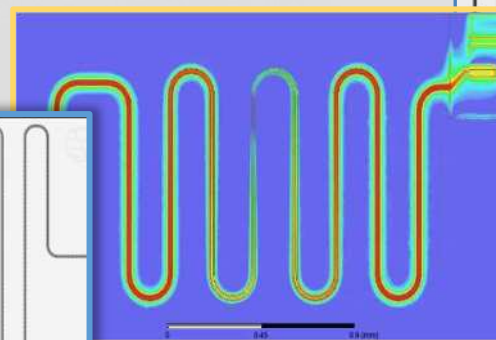
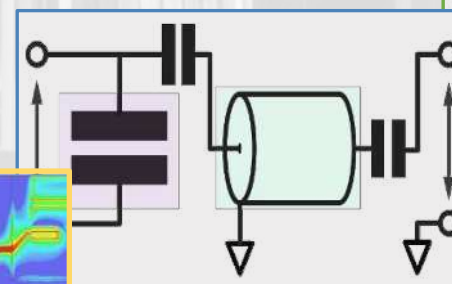
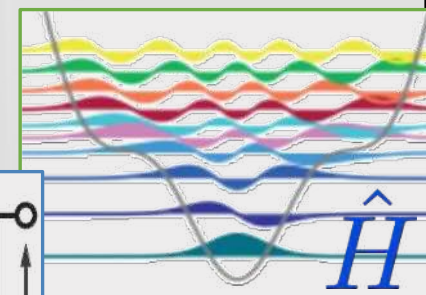
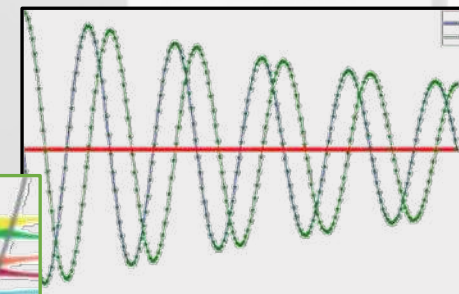
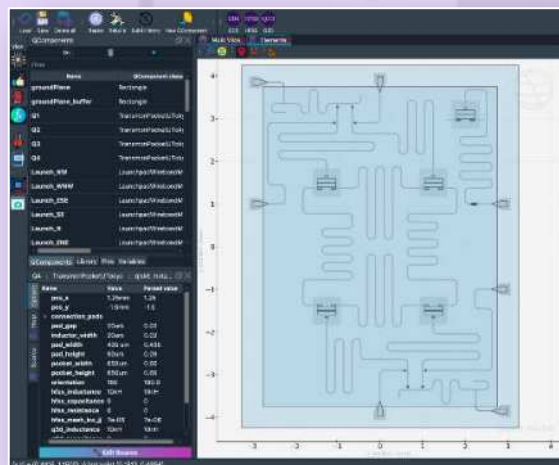
Quantum
Hamiltonian

Quantization
Methods

E&M
Simulation

Layout

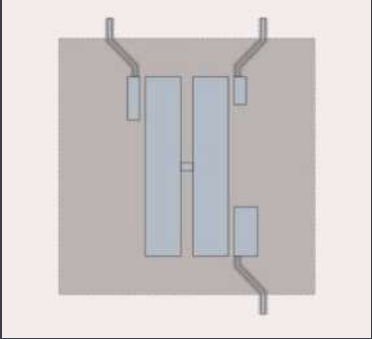
Fabrication



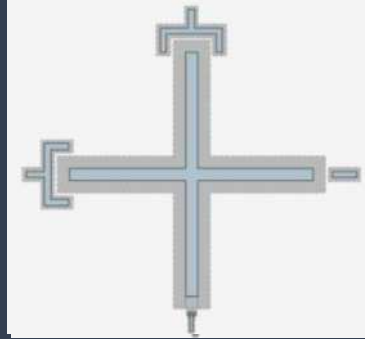
Device Library

Qubits

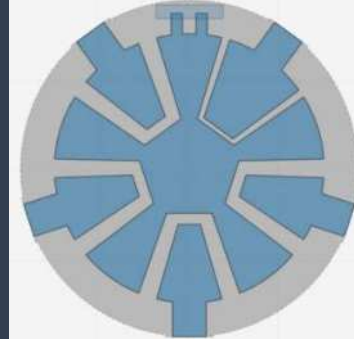
Resonators



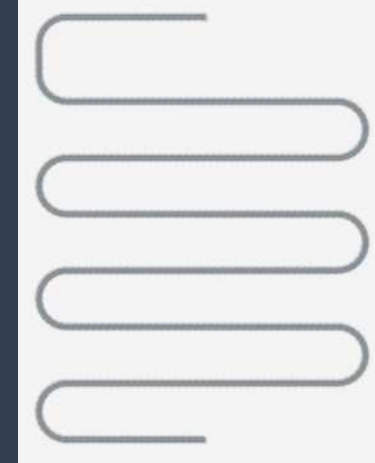
Single transmon-floating



Single transmon-grounded flux lines



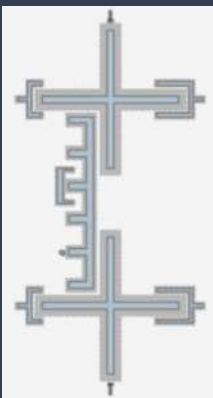
Star-shaped qubit



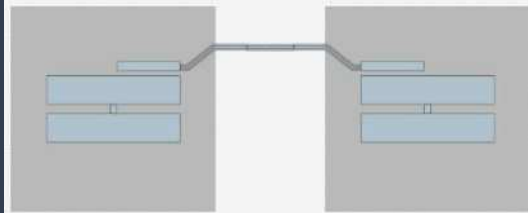
Simple coplanar waveguide (CPW) meander

Qubit Couplers

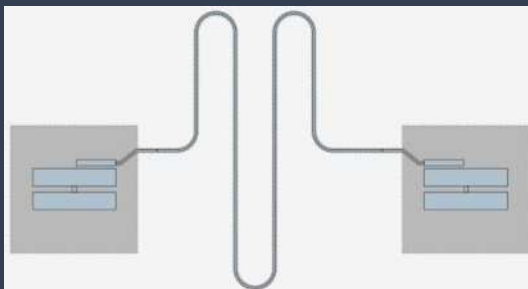
Input – Output Coupling



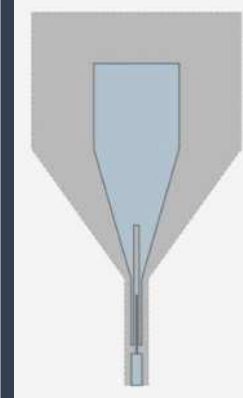
Tunable coupler (MIT)



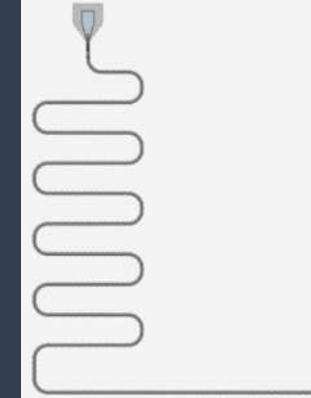
Direct coupler



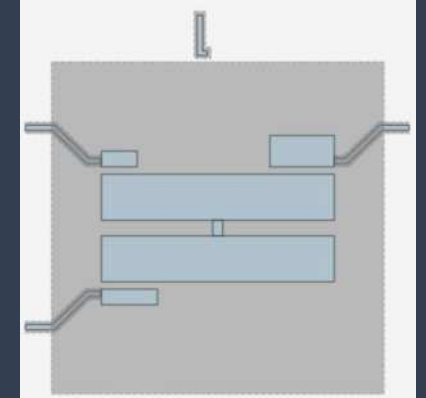
Bus resonator coupler



CPW launch pad



Readout line



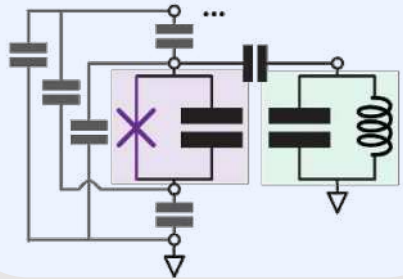
Charge line

Landscape of quantization methods

quasi-static

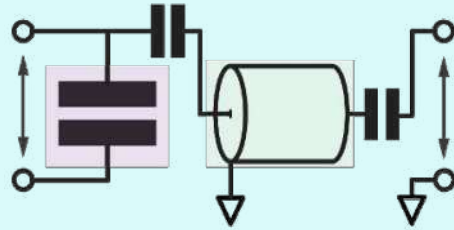
full-wave

lumped



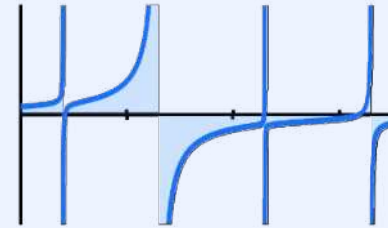
Yurke & Denker (1984), Devoret (1997), Burkard *et al.* (2004), Koch *et al.*...

quasi-lumped



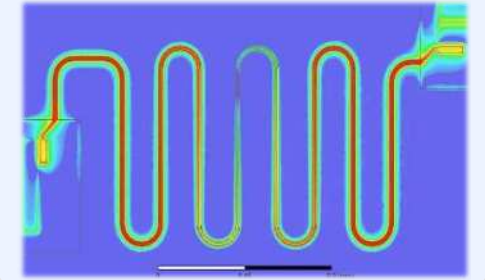
Malekakhlagh *et al.* (2017, 2019), Gely *et al.* (2019), Parra-Rodriguez *et al.* (2019), Minev *et al.* (2021), ...

impedance



Nigg *et al.* (2012), Bourassa *et al.* (2012), Solgun *et al.* (2014, 2015, 2017) ...

energy



Minev (2018)
Minev *et al.* (2020)

more speed

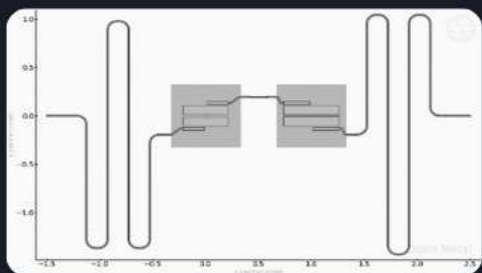
more information, accuracy



Qiskit Metal lumped model framework

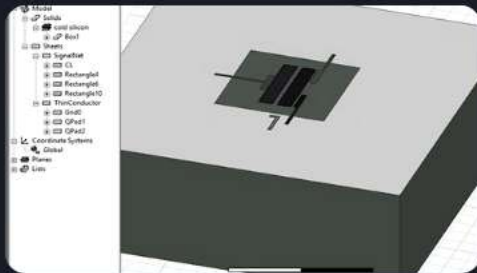
From layout to gates simulation in one notebook

Qiskit | quantum device design



Layout

Layout with Qiskit Metal's built-in EDA with either GUI or jupyter lab programitically



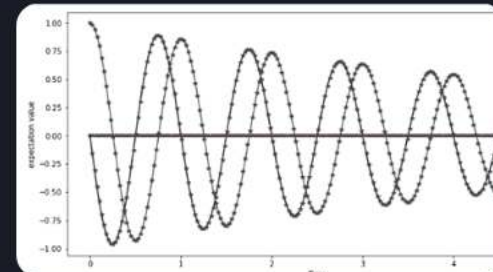
Electromagnetic simulation

Convenient interface with third-party EM simulation suites

$$\hat{H}_{\text{full}} = \hat{H}_0 + \sum_{n=1}^K \hat{H}_n + \sum_{n=0}^K \sum_{m=n+1}^K \hat{H}_{nm}$$

Quantum analysis

Lumped model analysis and Hamiltonian analysis



Pulse simulation

Convenient Hamiltonian interface with Qutip, Qiskit Dynamics, seQuencing for time evolution simulation

Resonators
Composite Bi-Partite Systems
Qubit Couplers
Input-Output Coupling
Small Quantum Chips
Design Flow

Libraries

All Quantum Devices

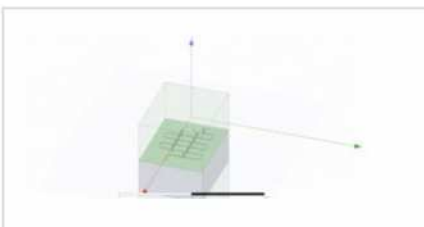
API References

Overview
QDesigns
QComponents
Analyses
QRenderes
Toolbox
QGeometry
GUI

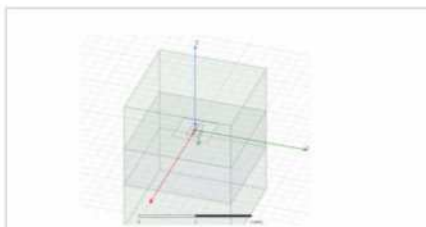
Code of Conduct



Tutorials: Quantum analysis library



Eigenmode and EPR



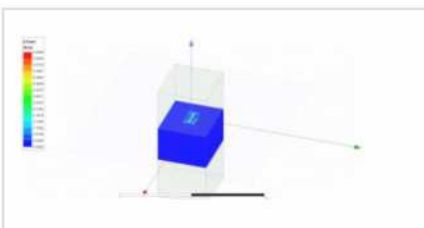
Impedance Analysis



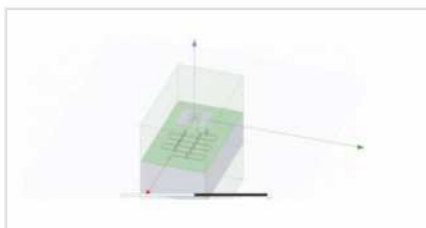
Capacitance Matrix and LOM Analysis



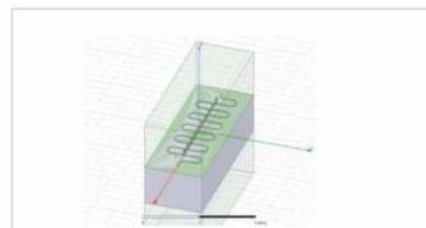
Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.



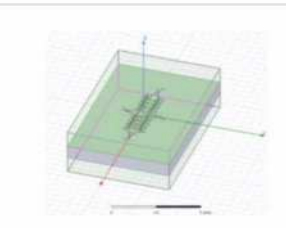
Analyzing and Tuning a Transmon Qubit



Analyzing a transmon and a resonator



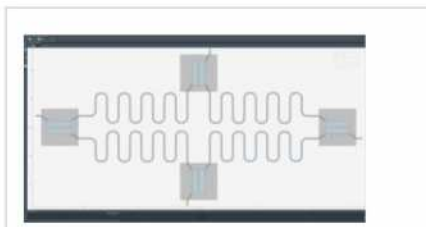
Analyzing a resonator



How to Render a Metal Design into Ansys



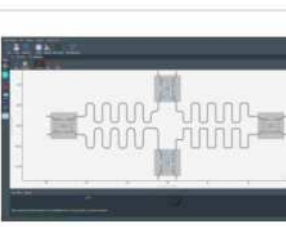
Driven Modal Simulation and S-Parameters



Sweep an option in QComponent, get capacitance matrix from Ansys Q3D and pyEPR



Sweep an option in QComponent, get Ansys HFSS eigenmode solution matrix.



How to Render a Metal Design into GDS

Resonators
Composite Bi-Partite Systems
Qubit Couplers
Input-Output Coupling
Small Quantum Chips
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API References

Overview

QDe

C

Tutorials: Quantum analysis library

Eigenmode and EPR

Impedance Analysis

Capacitance Matrix and LOM
Analysis

Sweep a option in QComponent,
get Ansys HFSS driven-modal
scatter matrix, impedance matrix,
and admittance matrix.

Eigenmode

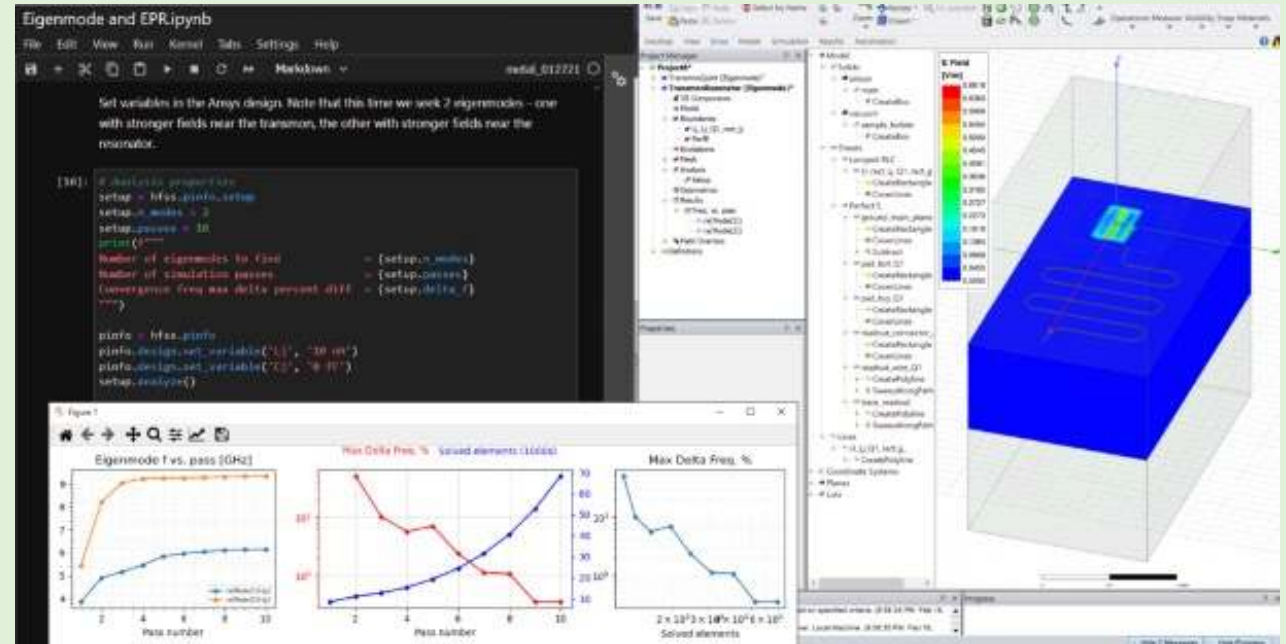
Energy-participation quantization of Josephson circuits

[Zlatko K. Minev](#) , [Zaki Leghtas](#), [Shantanu O. Mundhada](#), [Lysander Christakis](#), [Ioan M. Pop](#)
& [Michel H. Devoret](#)

[npj Quantum Information](#) **7**, Article number: 131 (2021) | [Cite this article](#)

Minev arXiv: 1902.10355

Minev arXiv: 2010.00620



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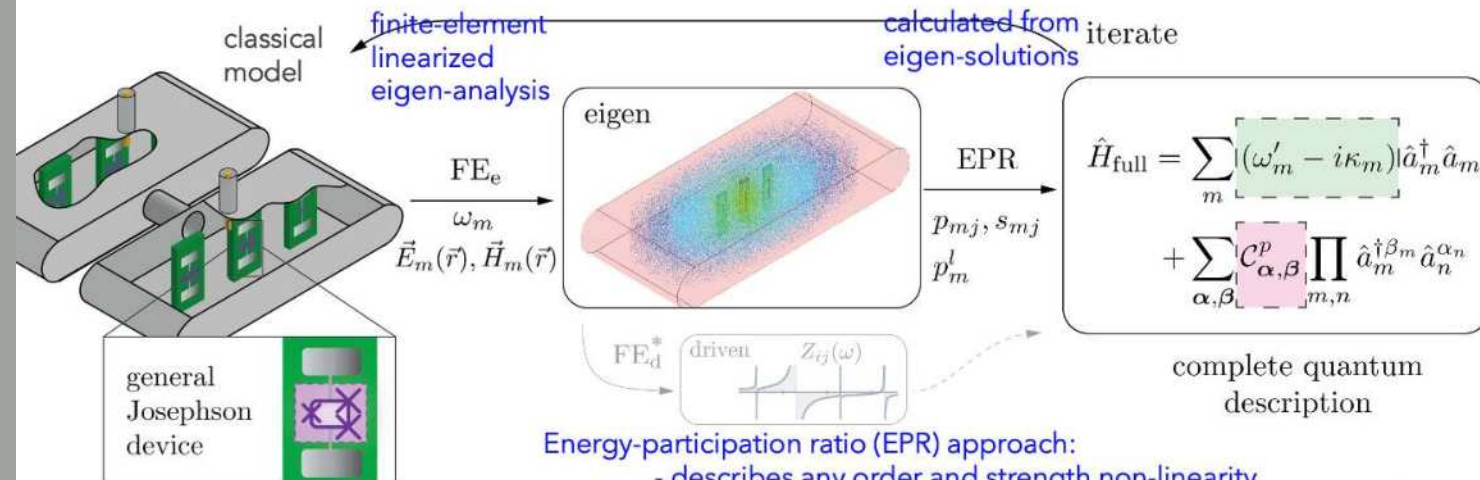
Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.

Energy-participation quantization of Josephson circuits

Zlatko K. Minev , Zaki Leghtas, Shantanu O. Mundhada, Lysander Christakis, Ioan M. Pop & Michel H. Devoret

[npj Quantum Information](#) **7**, Article number: 131 (2021) | [Cite this article](#)

Overview of energy-participation approach



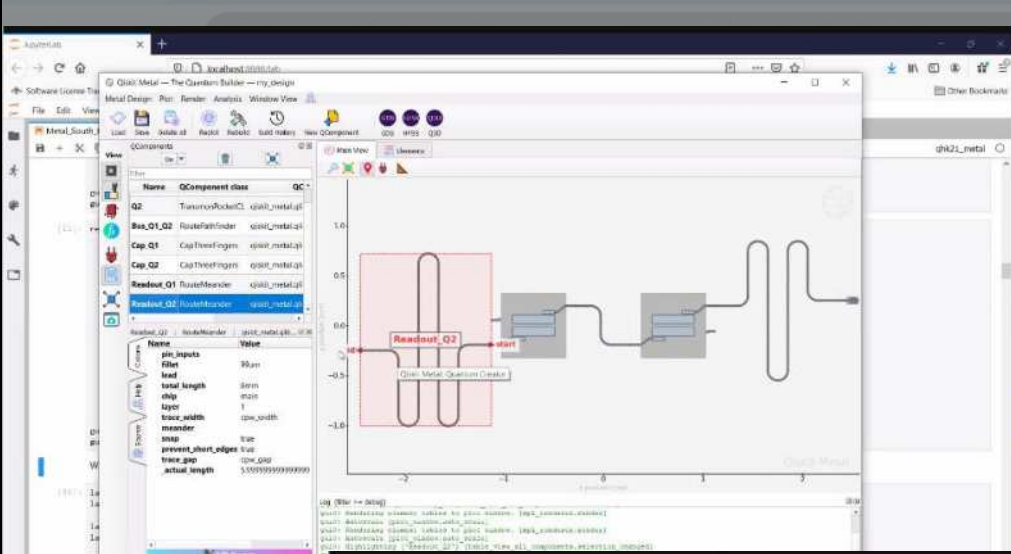
Energy-participation ratio (EPR) approach:

- describes any order and strength non-linearity
- describes arbitrary (composite) non-linear inductive devices
- first-principle derivation
- zero approximations (aside from truncation of modes)
- fully automated in python (github.com/zlatko-minev)

Practical limits: Fock and mode basis truncation due to computing power

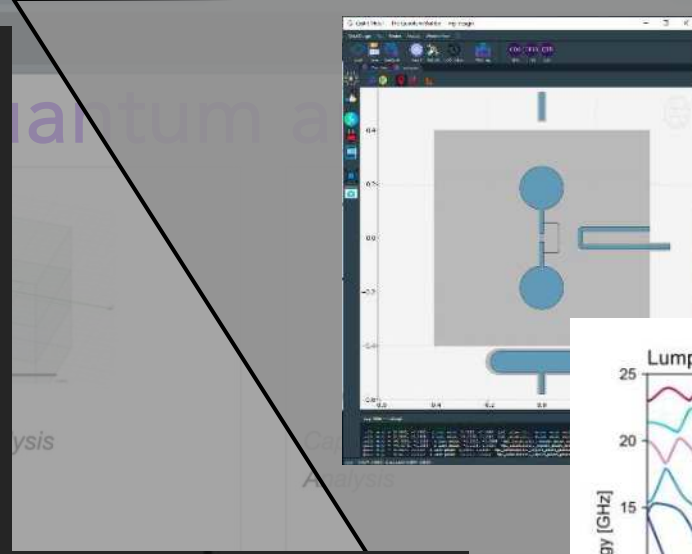
* Nigg, Paik, et al., PRL (2012),
Bourassa et al. (2012),
Solgun et al. (2014, 2015, 2017), ...

Minev arXiv: 1902.10355
Minev arXiv: 2010.00620

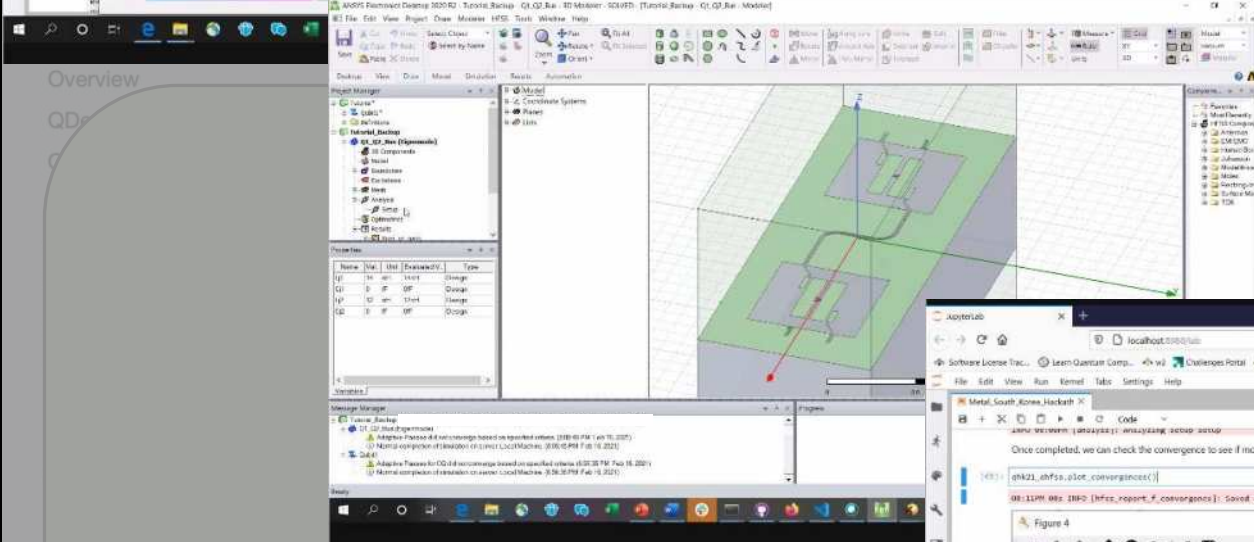


Zlatko Minev

Example: transmons



Example: fluxonium (Delft)



Zlatko Minev

Sam (Co-host)

Yuri Kobayashi (Co-host)

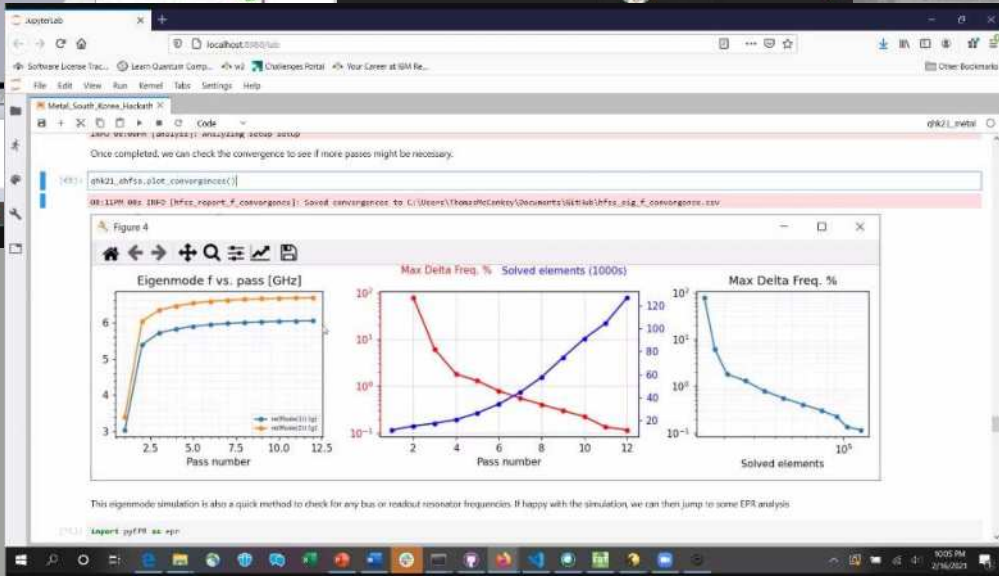
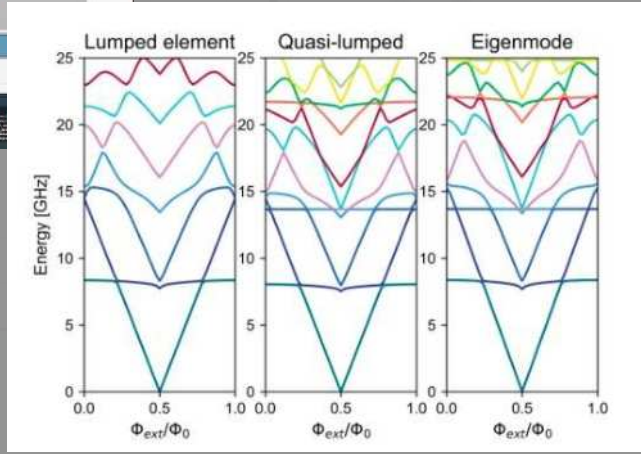
Atsushi Matsuo

BoSeong Kim

Byung Du Oh

Dongsin Kim

Eunchan Cho



Zlatko Minev

Sam (Co-host)

Yuri Kobayashi (Co-host)

Atsushi Matsuo

BoSeong Kim

Byung Du Oh

Dongsin Kim

Eunchan Cho

Gahyun Choi

goeun baek

Hanhee Paik

Heekang

Hiroshi Horii

Minev arXiv: 1902.10355

Minev arXiv: 2010.00620

Resonators
Composite Bi-Partite Systems
Qubit Couplers
Input-Output Coupling
Small Quantum Chips
Design Flow

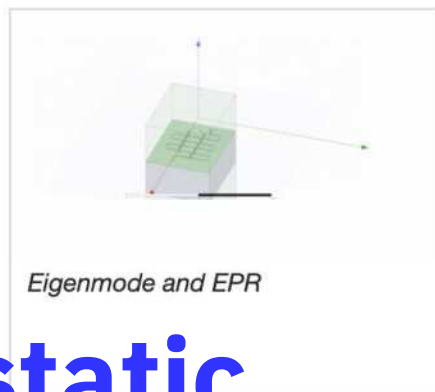
Libraries
All Quantum Devices

API References

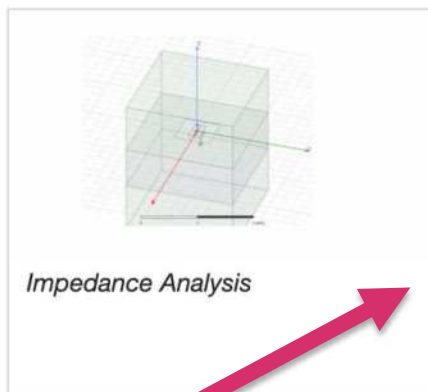
Overview
QDesigns
QComponents
Analyses
QRenderes
Toolbox
QGeometry
GUI

Code of Conduct

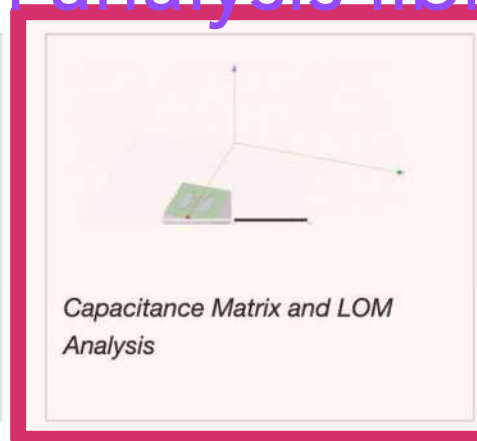
Tutorials: Quantum analysis library



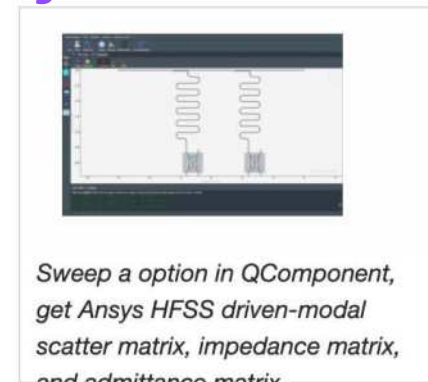
Eigenmode and EPR



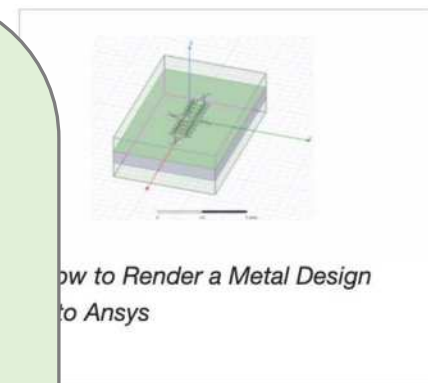
Impedance Analysis



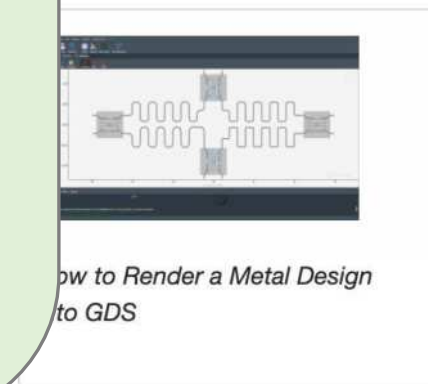
Capacitance Matrix and LOM Analysis



Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.



How to Render a Metal Design to Ansys



How to Render a Metal Design to GDS

Quasi-static



Cornell University

arXiv.org > quant-ph > arXiv:2103.10344

Quantum Physics

Circuit quantum electrodynamics (cQED) with modular quasi-lumped models

Zlatko K. Minev,^{*} Thomas G. McConkey, Maika Takita, Antonio Corcoles, and Jay M. Gambetta
IBM Quantum, IBM T.J. Watson Research Center, Yorktown Heights, US

General capacitive analysis code
map to known building blocks: e.g.,
transmon, fluxonium, zero-pi, etc.

Tutorials: Quantum analysis library

Simple example

API References

Overview

QDesigns

QComponents

Analyses

QRenderes

Toolbox

QGeometry

GUI

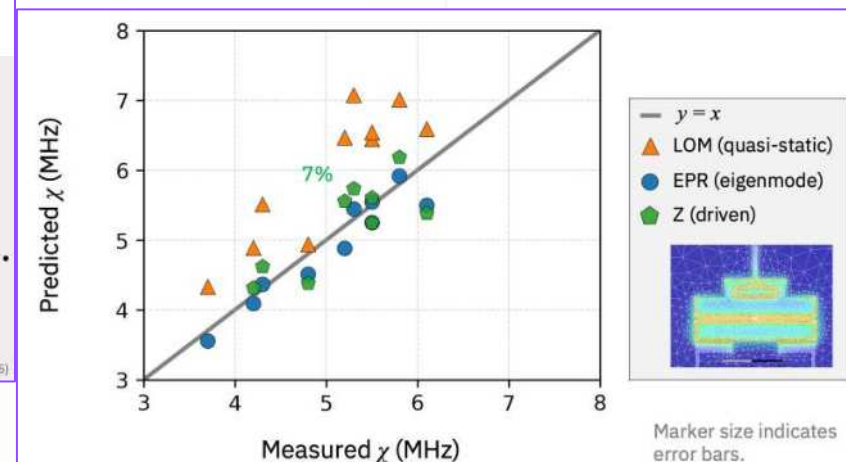
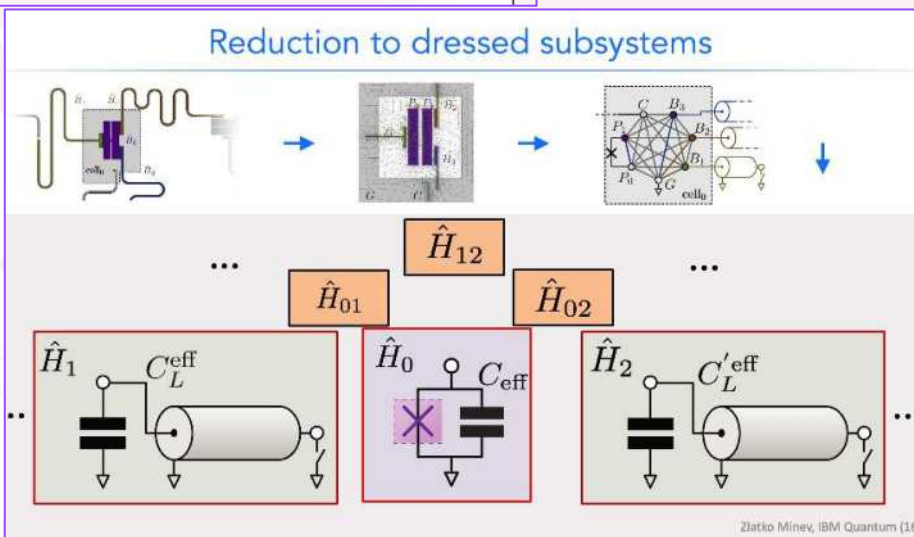
Code of Conduct

arXiv.org > quant-ph > arXiv:1608.03366v2 [quant-ph]

Quantum Physics

Circuit quantum ele

Zlatko K. Mineev[*] Thor
IBM Quan



load transmon cell Q3d simulation results

```
path1 = './Q1_TwoTransmon_CapMatrix.txt'
ta_mat, _ = load_q3d_capacitance_matrix(path1, _disp=False)

path2 = './Q2_TwoTransmon_CapMatrix.txt'
tb_mat, _ = load_q3d_capacitance_matrix(path2, _disp=False)
```

Create LOM cells from capacitance matrices

```
# cell 1: transmon Alice cell

opt1 = dict(
    node_rename = {'coupler_connector_pad_Q1': 'coupling', 'readout_connector_pad_Q1': 'readout_alice'},
    cap_mat = ta_mat,
    ind_dict = {'pad_top_Q1': 'pad_bot_Q1': 10},
    j1_dict = {'pad_top_Q1': 'pad_bot_Q1': 1},
    cj_dict = {'pad_top_Q1': 'pad_bot_Q1': 2},
)
cell_1 = Cell(opt1)
```

```
# cell 2: transmon Bob cell

opt2 = dict(
    node_rename = {'coupler_connector_pad_Q2': 'coupling', 'readout_connector_pad_Q2': 'readout_bob'},
    cap_mat = tb_mat,
    ind_dict = {'pad_top_Q2': 'pad_bot_Q2': 12},
    j1_dict = {'pad_top_Q2': 'pad_bot_Q2': 1},
    cj_dict = {'pad_top_Q2': 'pad_bot_Q2': 2},
)
cell_2 = Cell(opt2)
```

Make subsystems

```
# subsystem 1: transmon Alice
transmon_alice = Subsystem(name='transmon_alice', sys_type='TRANSMON', nodes=['j1'])

# subsystem 2: transmon Bob
transmon_bob = Subsystem(name='transmon_bob', sys_type='TRANSMON', nodes=['j2'])

# subsystem 3: Alice readout resonator
q_opts = dict(
    f_res = 8, # resonator dressed frequency in GHz
    Z0 = 50, # characteristic impedance in Ohm
    vp = 0.404314 * c_light # phase velocity
)
res_alice = Subsystem(name='readout_alice', sys_type='TL_RESONATOR', nodes=['readout_alice'], q_opts=q_opts)

# subsystem 4: Bob readout resonator
q_opts = dict(
    f_res = 7.6, # resonator dressed frequency in GHz
    Z0 = 50, # characteristic impedance in Ohm
    vp = 0.404314 * c_light # phase velocity
)
res_bob = Subsystem(name='readout_bob', sys_type='TL_RESONATOR', nodes=['readout_bob'], q_opts=q_opts)
```

Creat the composite system from the cells and the subsystems

```
composite_sys = CompositeSystem(
    subsystems=[transmon_alice, transmon_bob, res_alice, res_bob],
    cells=[cell_1, cell_2],
    grd_node='ground_main_plane',
    nodes_force_keep=['readout_alice', 'readout_bob']
)
```

Generate the hilbertspace from the composite system

```
hilbertspace = composite_sys.create_hilbertspace()
hilbertspace = composite_sys.add_interaction()
```

Print the results

```
hamiltonian_results = composite_sys.hamiltonian_results(hilbertspace, evals_count=30)
```

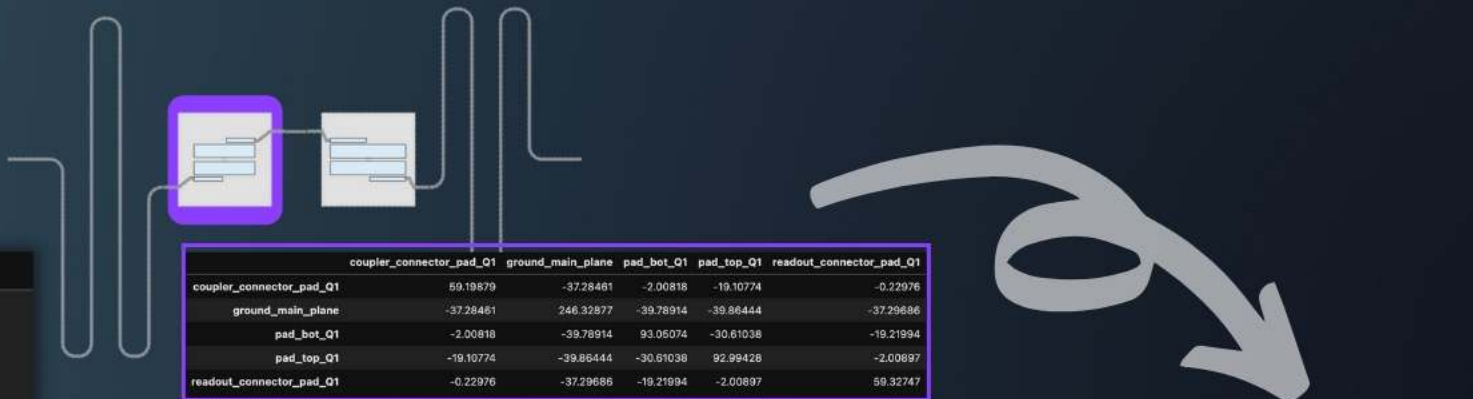
Finished eigensystem.

system frequencies in GHz:

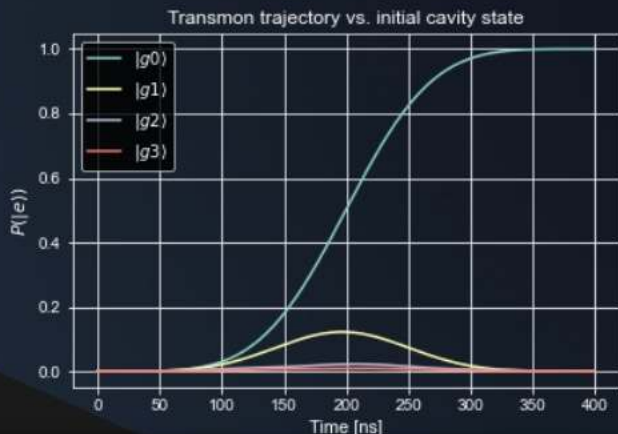
```
{'transmon_alice': 6.053360688806868, 'transmon_bob': 4.7989883222888094, 'readout_alice': 8.0090548
```

Chi matrices in MHz

	transmon_alice	transmon_bob	readout_alice	readout_bob
transmon_alice	-353.239816	-0.542895	-4.132854	-0.003120
transmon_bob	-0.542895	-263.940098	-0.001154	-1.460416
readout_alice	-4.132854	-0.001154	4.283111	-0.000017
readout_bob	-0.003120	-1.460416	-0.000017	3.829744



	coupler_connector_pad_Q1	ground_main_plane	pad_bot_Q1	pad_top_Q1	readout_connector_pad_Q1
coupler_connector_pad_Q1	50.19870	-37.28461	-2.00818	-19.10774	-0.22976
ground_main_plane	-37.28461	246.32877	-39.78914	-39.86444	-37.29686
pad_bot_Q1	-2.00818	-39.78914	93.05074	-30.61038	-18.21994
pad_top_Q1	-19.10774	-39.86444	-30.61038	82.99428	-2.00897
readout_connector_pad_Q1	-0.22976	-37.29686	-18.21994	-2.00897	59.32747



From layout to time evolution simulation in one simple notebook

Time evolution simulation

```
system = lom_composite_sys_to_seq_sys(composite_sys, hilbertspace, levels=[3, 3, 10, 10])
alice = system.modes[1]
```

Finished eigensystem.

```
selective_sigma = 100 # ns
```

```
# tune selective qubit pulse using Rabi
with system.use_modes([alice]):
    with alice.temporarily_set(gaussian_pulse_sigma=selective_sigma):
        selective_qubit_amp = tune_rabi(
            system, system.fock(transmon_alice=0, transmon_bob=0, readout_alice=0, readout_bob=0)
        )
```

```
100%|██████████| 51/51 [00:01<00:00, 41.64it/s]
```

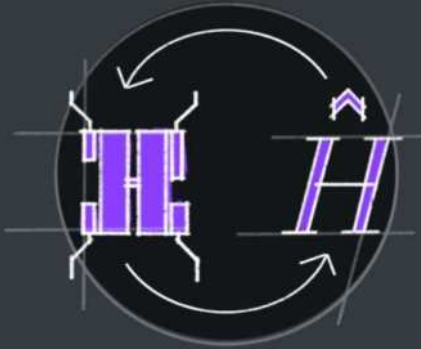
```
def selective_rotation(qubit, angle, phase=0, detune=0, sigma=selective_sigma):
    with qubit.gaussian_pulse.temporarily_set(sigma=sigma, amp=selective_qubit_amp):
        qubit.rotate(np.pi, phase, detune=detune)
```

Apply a selective pi pulse that is resonant with the qubit when the cavity is in $|0\rangle$.

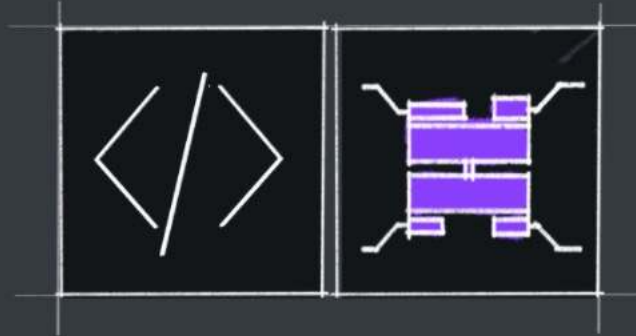
```
init_states = [
    (f'$|g{n}\rangle$', system.fock(transmon_alice=0, readout_alice=n)) for n in range(4)
]
```

Qiskit Metal: Key features and content

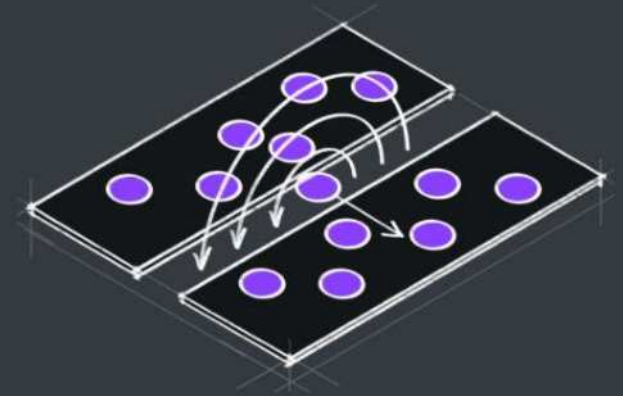
End-to-end automation



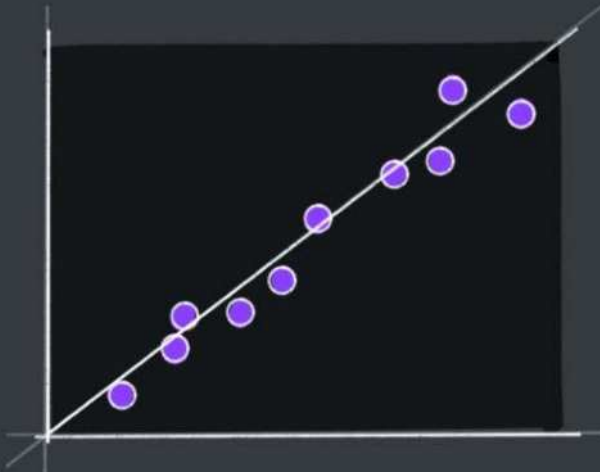
Flexible & extensible



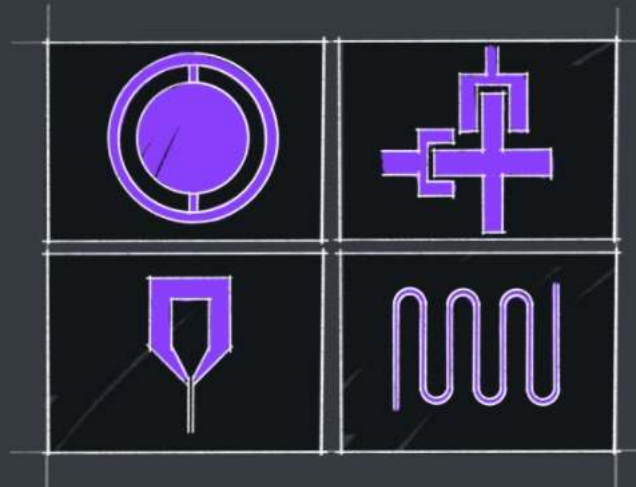
Light-weight interoperability



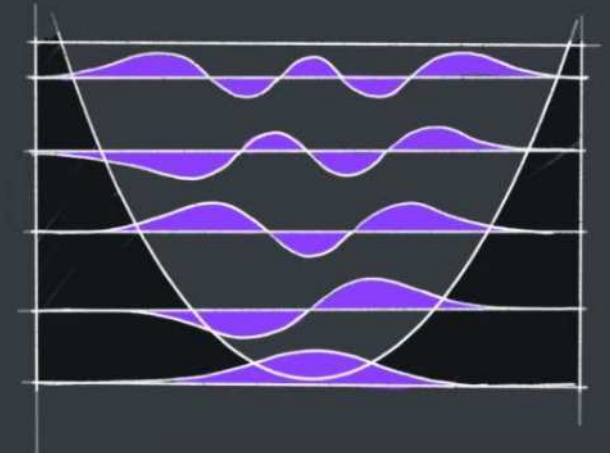
Experimentally tested



Library of components



Cutting edge resources



The buzz...

Forbes

You Don't Have To Be A Rocket (Or Quantum) Scientist To Design A Quantum Computer Chip Using IBM's New Tool Called Qiskit Metal

Paul Smith-Goodson Contributor
Moor Insights and Strategy Contributor Group ©
Cloud
Analyst-in-residence, Quantum Computing



Qiskit | quantum device design

Qiskit Metal | quantum device design is now open

IBM Qiskit IBM

Intuitively, almost everyone can appreciate how difficult and knowledge-intensive it is to design, develop, analyze, and simulate a quantum computer chip. Without years

Public launch of Qiskit Metal (2021)

“Considering all its advantages, *Qiskit Metal* should be a clear long-term winner for IBM and the quantum community.”

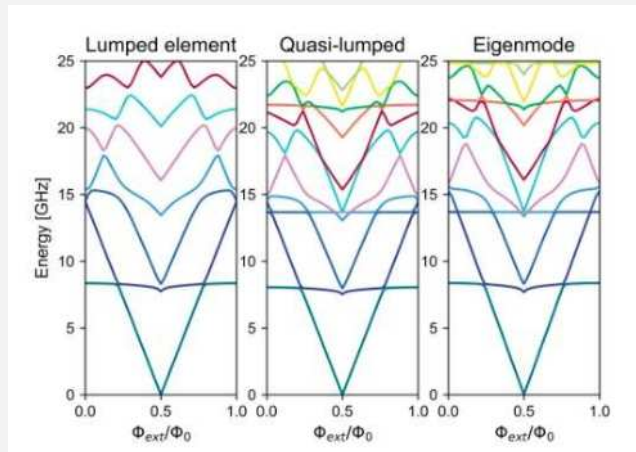
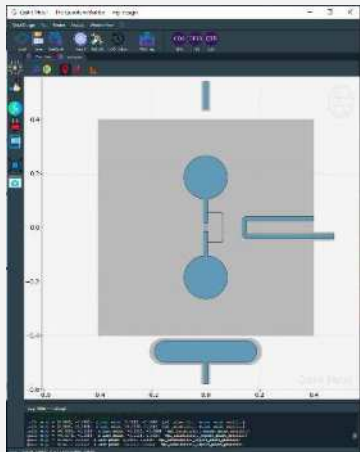
- Forbes

QuTech / Technische Universiteit Delft

Figen Yilmaz-Andersen Lab: Fluxonium chip design

Roald van den Boogaart-Andersen Lab: Fluxonium spectrum (eigenmode = pyEPR) to show how they have extended the analysis methods to be able to handle fluxonium qubits

Sara Buhktari-Andersen Lab: Fabricated resonator design being measured



3.1. Final design

The final design consists of an S-shaped feedline routed between two launchpads. Eight resonators are capacitively coupled to this feedline. Every component on this 9 x 9 mm device is implemented by a coplanar waveguide transmission line. The schematic representations of the full device and single resonator geometry are depicted in Figure 6.

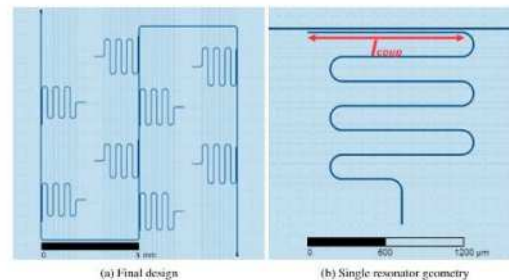
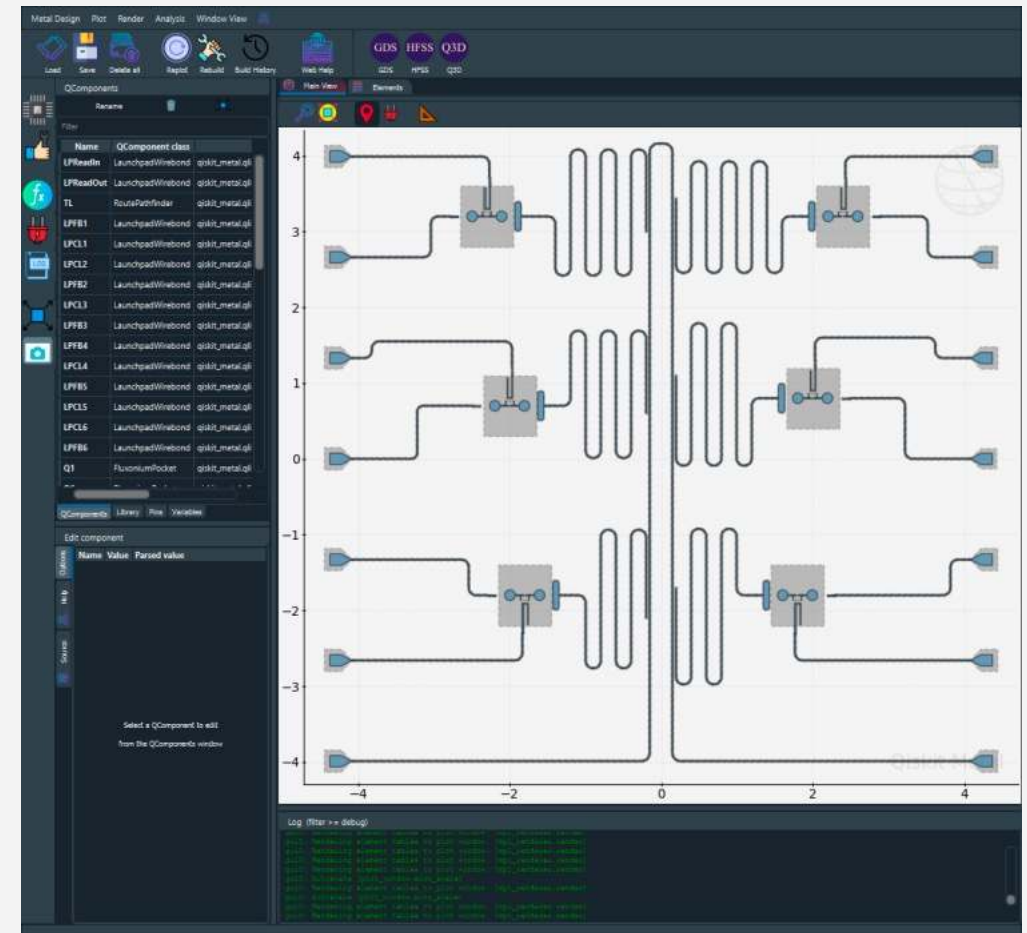


Figure 6: Schematic representations of the final design and single resonator geometry. Each resonator is capacitively coupled to the feedline. The resonator coupling length is denoted by l_{coup} .

“Qiskit-Metal is the best tool if one studies superconducting qubits”
-- Figen Yilmaz

QuTech / Technische Universiteit Delft

Intermediate chip design. Quantum chip fabricated, cooled down, measured, reporting at APS MM23



Zlatko Minev, IBM Quantum

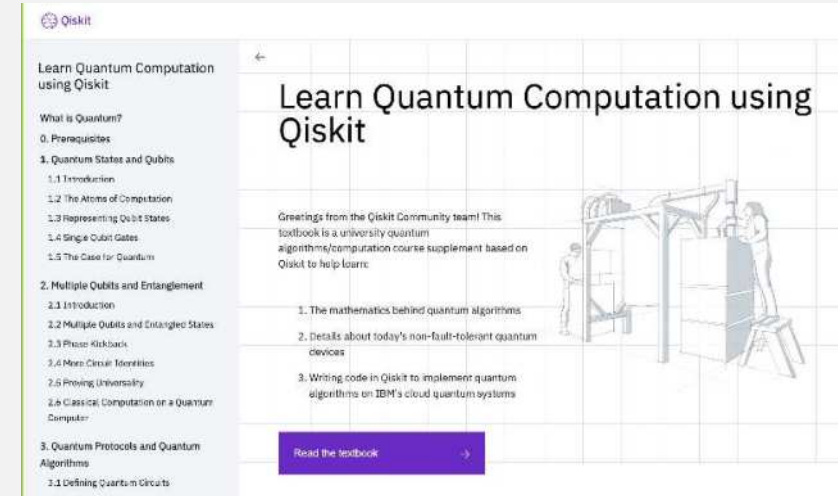
Ways to learn more

Qiskit Global Summer School

Qiskit  YouTube (Lec. 16-21 by Z. Minev)

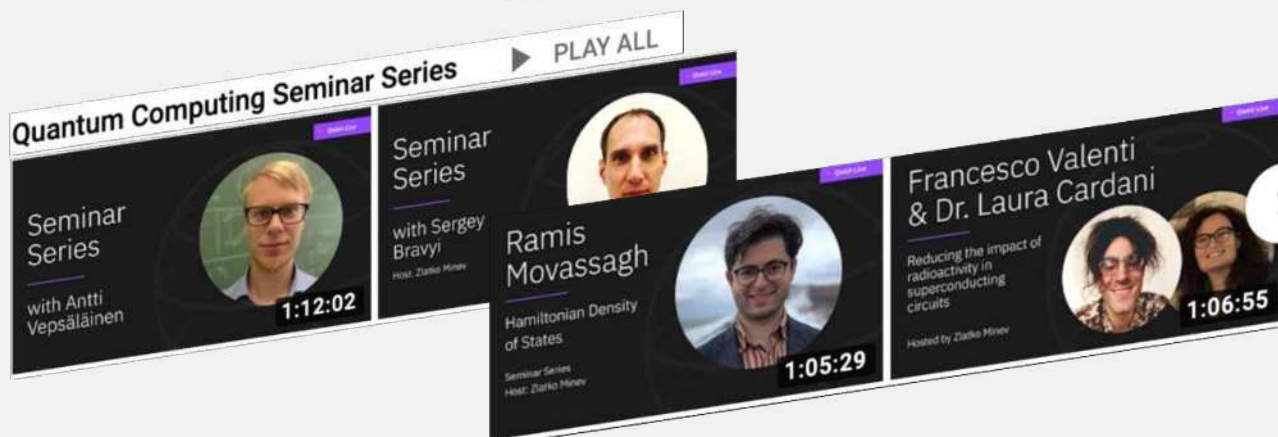


qiskit.org/textbook



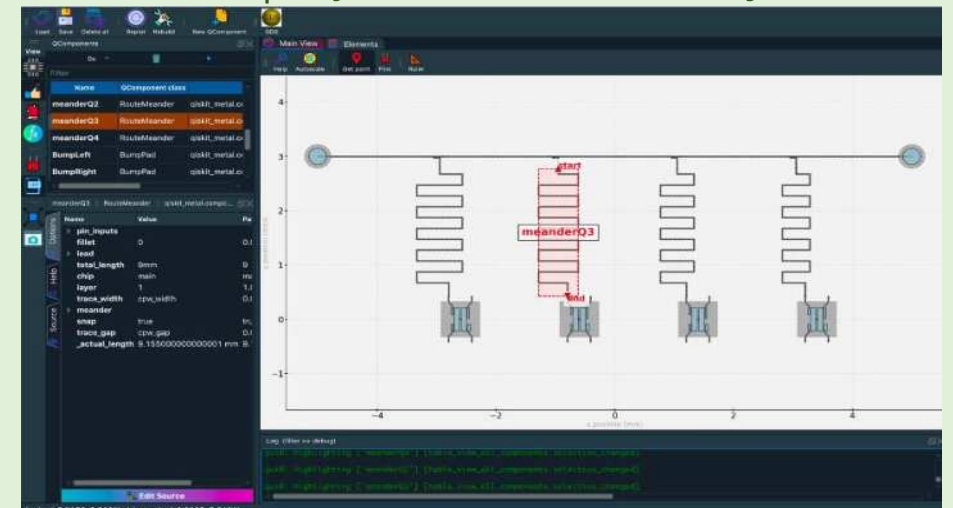
Quantum Information Seminar Series

(Qiskit  YouTube Host: Z. Minev)



Qiskit Metal — PowerUp

Python-based project started & led by Z. Minev



The important thing is not to stop questioning.
Curiosity has its own reason for existence.

One cannot help but be in awe when he
contemplates the mysteries of eternity, of life, of the
marvelous structure of reality.

It is enough if one tries merely to comprehend a
little of this mystery each day.

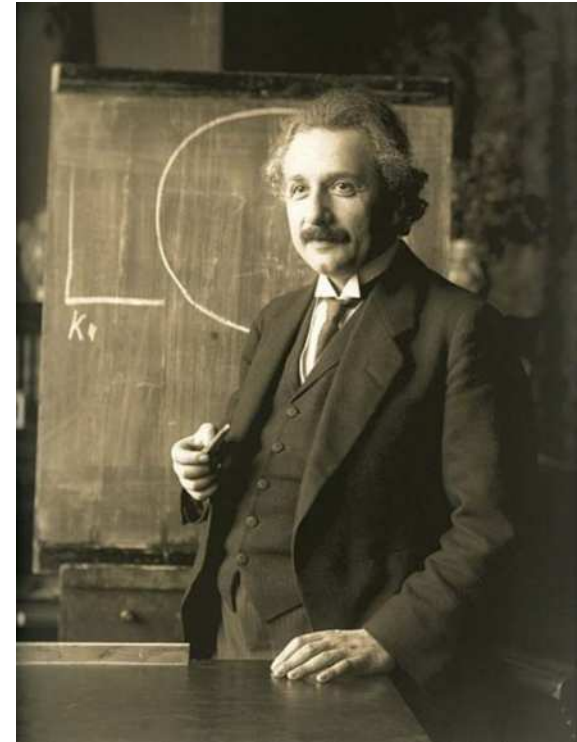


Photo: F. Schmutzer

Albert Einstein



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Open Questions in Quantum Computing v0.1

1. **Where does the power of quantum computing come from?**

useful to answer in order to design new quantum algorithms, compare to classical, and to find quantum advantages D/W: I. Deutsch, ...

2. **What will quantum computers teach us about fundamental quantum physics?**

predictability, collapse, tunneling time, many body scars, many worlds, ... D/W: A. Steinberg, I. Siddiqi, R. Gutierrez-Jauregui, P. Zoller, M. Lukin, ...

3. **How can we efficiently extract information from quantum data (bridge quantum-to-classical boundary)?**

E.g., learn many-body properties of large quantum systems, noise, channels. Challenges: Need to overcome fundamental no-go theorems in the bridge from quantum to classical interface, exponential scaling for quantum properties, validation, noise plague, ...

D/W: L. Jiang, J. Preskill, P. Zoller, M. Lukin, ...

4. **Is it useful to build a quantum computer with 1% error?**

What do we do with noisy pre-fault-tolerant quantum processors? D/W: J. Martinis

5. **How to craft conventionally forbidden but useful quantum operations?**

such as inverse of a large noise map; e.g., quasi-probability, ancilla-assisted, ... D/W: with K. Temme, S. Worner, D. Sutter, ...

6. **How to overcome noise in real quantum processors?**

Unmonitored systems, error mitigation, novel techniques, path to 100 qubits x 400 depth ...

7. **What's beyond the thermodynamic limit and ETH in many-body physics?**

Qs stimulated by quantum computers; e.g., MITP, MBL probes. D/W: M. Lukin, P. Zoller, V. Khemani, H. Pichler, S. Gopalakrishnan, M. Hafezi, A. Seif, D. Wang, ...

8. **What tasks can quantum outperform classical in that we can demonstrate in the next 1-3 years?**

Thank you!

Life & lessons as a scientist
in “real life”



Zlatko K. Minev, Ph.D.

IBM Quantum, T.J. Watson Research Center, Yorktown Heights, NY



@zlatko_minev



zlatko-minev.com

success **noun**

suc·cess | \ sək-'ses  \

Definition of *success*



the attainment of wealth, favor, or eminence

“Success is peace of mind that is the direct result of self-satisfaction in knowing you did your best to become the best that you are capable of becoming.”

Coach John Wooden
(1910–2010)

